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TITLE:

Energy management system application program interface (EMS-API) - Part 600-1: Common Grid Model Exchange Specification (CGMES) - Structure and rules

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Foreword

The purpose of the Common Grid Model Exchange Specification (CGMES) is to define the interface between Transmission System Operators (TSO) software in order to exchange power system modelling information as required by the European Network of Transmission System Operators for Electricity (ENTSO-E) and TSO business processes.

The CGMES is used as a baseline exchange standard for the implementation of the Common Grid Model (CGM) methodologies in accordance with the requirements for the implementation of various European network codes and guidelines. The CGMES applies to applications dealing with power system data management, as well as applications supporting the following analyses:

- load flow and contingency analyses,
- short circuit calculations,
- market information and transparency,
- capacity calculation for capacity allocation and congestion management, and
- dynamic security assessment.

The conformity of the applications used for operational and system development exchanges with the CGMES is crucial for the needed interoperability of these applications. ENTSO-E therefore developed and approved the CGMES Conformity Assessment Framework as the guiding principles for assessing applications' CGMES conformity. Based on those principles. This technical specification relies on the CGMES Conformity Assessment Process operated by ENTSO-E in order to ensure that the CGMES is properly implemented by suppliers of the applications used by TSOs.

Document history

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Introduction

1 Scope

This technical specification on the CGMES defines the main rules and requirements related to the CGMES and which are mandatory for achieving interoperability with the CGMES and for satisfying the business processes. In the document requirements are indicated as such in a tabular format. Some descriptions are merely used for clarification and are marked “Informational”.

The profiles which belong to CGMES are defined in a separate document. The related technical information and documentation (i.e. RDFS, OCL, XMI and HTML) needed for the implementation of the CGMES, which is not copyrighted by either IEC or CENELEC, is available at the ENTSO-E web site.

The CGMES is defined using information on the Common Information Model (CIM) available in the public domain.

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

The CGMES is a superset of the former ENTSO-E CIM based data exchange standard (Profile 1) which was based on CIM14 (UML14v02) and has been used for certain network models exchanges since 2009. The CGMES reflects TSO requirements (as known by 2014) for accurate modelling of the ENTSO-E area for power flow, short circuit, and dynamics applications whilst also allowing for the exchange of any diagram layouts including GIS data of a grid model.

Next editions of this technical specification will be released to describe following CGMES versions which will be reflecting the additional requirements due to European network codes or guidelines.

2 Normative references

The CGMES is based on the CIM 16 (UML16v29). Once the following IEC CIM standards are approved the CGMES would rely on them:

- IEC 61970-552 : CIM XML Model Exchange Format
- IEC 61970-301 : Common Information Model (CIM) Base
- IEC 61970-302 : Common Information Model (CIM) for Dynamics Specification
- IEC 61970-452 : CIM Static Transmission Network Model Profiles
- IEC 61970-453 : Diagram Layout Profile
- IEC 61970-456 : Solved Power System State Profiles
- IEC 61970-457 : Common Information Model (CIM) for Dynamics Profile
- IEC 61970-501 : Common Information Model Resource Description Framework (CIM RDF) schema
- IEC 61968-4 : Application integration at electric utilities – System interfaces for distribution management - Part 4: Interfaces for records and asset management.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply. For definitions which are not specified in the CGMES the definitions in the IEC CIM related standards shall be applied.

199 **3.1**
200 **Common Grid Model Exchange Standard (CGMES)**
201 This is an ENTSO-E standard used for the exchange of power system models between TSOs for the purpose
202 of performing bilateral, regional or pan-European studies in the frame of TYNDP or TSOs' projects. It is based
203 on IEC CIM Standards and further extended to meet Network Codes' and projects' requirements. The
204 standard defines a set of data model exchange profiles.

205 **3.2**
206 **Profile**
207 This refers to the uniquely named subset of classes, associations and attributes needed to accomplish a
208 specific type of interface and based upon a canonical model. The term profile may be used to define either
209 the semantic model for an instance data payload or the syntactic schema for an instance data payload. A
210 profile may be expressed in XSD, RDF, and/or OWL files. An instance data conforming to a profile can be
211 tested in exchanges between applications. A profile is necessary in order to "use" the canonical model.

212 **3.3**
213 **CIM Extension**
214 A collection of classes, attributes and associations, which extend the standard IEC CIM model in order to
215 cover use cases not currently supported by IEC standards, and which are not considered to be international
216 use cases or are covered by a later version of the standard which is not yet supported.

217 **3.4**
218 **ENTSO-E Extension**
219 CIM Extension, specifically managed by ENTSO-E.

220 **3.5**
221 **Boundary set**
222 A boundary set contains all boundary points necessary for a given grid model exchange. A boundary set can
223 have different coverage depending on the requirements of the common grid model exchange. A complete
224 boundary set is necessary to assemble a pan-European power system model.

225 **3.6**
226 **Boundary point (BP)**
227 Boundary point is a connection point between two Model Authority Sets (MAS). A Boundary point could be a
228 ConnectivityNode or a TopologicalNode placed on a tie-line or in a substation. A Boundary point must be
229 contained in a Boundary Set and must not be contained in the MAS of a TSO. A Boundary point is referenced
230 by Terminals in the MAS of a TSO. ConnectivityNode and TopologicalNode are terms specified in IEC CIM
231 standards. If a Boundary point is placed on a tie-line, the term X-Node is often used instead of Boundary point.
232 X-Node is therefore a specific type of Boundary point.

233 **3.7**
234 **Abbreviations**

IEC	The International Electrotechnical Commission, headquartered in Geneva
TSO	Transmission System Operator
ENTSO-E	European Network of Transmission System Operators for Electricity. ENTSO-E has 41 TSO members.
MRID	CIM Master Resource Identifier
CIM	Common Information Model (electricity)
CGMES	Common Grid Model Exchange Standard
MAS	Model Authority Set
IOP	Interoperability Test

RDF	Resource Description Framework
EQ_BD	Boundary equipment profile or instance file
TP_BD	Boundary topology profile or instance file
EQ	Equipment profile or instance file
TP	Topology profile or instance file
SSH	Steady State Hypothesis profile or instance file
SV	State Variables profile or instance file
DL	Diagram Layout profile or instance file
GL	Geographical Location profile or instance file
DY	Dynamics profile or instance file
BP	Boundary point

4 Exchange process

There are various levels in which the exchange of power system data/models is necessary. A pan-European model exchange level covers the territory of all TSOs. Regional model exchanges can be realised between different TSOs in one or more synchronous areas. A model exchange on the national level includes interfaces between TSOs and DSOs, as well as between different DSOs.

The purpose of model exchanges is not only to exchange the data from one authority to another but also to satisfy the ultimate goal, namely to perform common studies using shared data. All parties involved in the process should be able to perform the same types of studies and be able to share project tasks between different parties which are using different power system analysis applications. Indeed, the interoperability between different applications used in the exchange process is therefore crucial in both reaching seamless data exchange and obtaining comparable study results when using this data.

The CGMES covers these ENTSO-E and TSO business processes by defining the following main types of exchanges valid for a particular study or process:

- Exchange of Boundary set: An exchange of a Boundary Set is necessary to prepare an exchange of an internal TSO model and to assemble a common grid model. The latest information on Boundary Sets covering pan-European area is available to TSOs and maintained in the ENTSO-E Network Modelling Database (NMD) where all TSOs negotiate and agree on the boundary information.
- Exchange of an Internal TSO model: A number of business processes require each TSO to provide models of its internal territory. To describe its internal territory in a single stand-alone exchange, a TSO is treated as a single model authority set and shall be able to exchange all profiles defined in the CGMES. The TSO prepared its internal model in such a way that it is easily and unambiguously combined with other TSO internal models to make up complete models for analytical purposes. This type of exchange can also be applied for the interface between a TSO and a DSO, where models covering transmission or distribution parts of the power system can be exchanged based on a mutual agreement between the TSOs and the DSOs. In this case, and if a TSO requests a DSO model, the DSO would provide its model in accordance with CGMES definitions which might be extended by the TSO requesting this type of exchange.

- Exchange of a Common grid model: A common grid model refers to the concept of having one model which can be used for multiple purposes. The specification describes what is needed to create an assembly of multiple TSOs Individual Grid Model (IGM) of their responsible territory into a regional or pan-European model. Different business processes will require specific implementation of the profiles part of the CGMES and the exchange of respective instance files to meet interoperability inside the business process. The Common Grid Model meta-model description will ensure interoperability across the business process.

ENTSO-E and TSO business processes (e.g. system development planning, protection planning, operational planning, operation, fault study/simulation, market operation, etc.) are, of course, more complex than these operations, but what is important to note is that all processes are supported using only these basic kinds of interoperation.

Note that each power system model in CIM normally consists of multiple datasets (instance files) as defined in IEC CIM Standards and further specified by CGMES.

The CGMES supports a node-breaker and a bus-branch model exchanges. Moving forward the procedures of the model exchanges using the CGMES, it is expected that equipment and steady state hypothesis data (EQ and SSH instance files) will be the input source data for all processes. This type of model should be the fully detailed model with all disconnectors/breakers, etc. Any configuration changes are made by changing switch statuses.

ID	Specification	Type
EXCH1.	The CGMES defines equipment and steady state hypothesis profiles as an input, meaning that all results, whether topology or state variables profiles data, must refer to the equipment and steady state hypothesis objects. Therefore, in the case that both equipment and steady state hypothesis instance files are available, there is no need to exchange topology or state variables instance files in order to obtain a load flow.	Requirement
EXCH2.	For node-breaker model exchanges the TopologicalNodes represent the output from a topology processing on the detailed input source operational data. These can be optionally exchanged to be used by tools which have an interest in the computed buses.	Information
EXCH3.	For node-breaker model exchanges mRID (rdfIDs in serialisation) of the TopologicalNodes are not persistent.	Information
EXCH4.	For node-breaker model exchanges a topology instance file is not exchanged using a difference file.	Requirement
EXCH5.	For bus-branch model exchanges the TopologicalNodes must be persistent.	Requirement
EXCH6.	If a contingency list is exchanged belonging to the model exchanged in bus-branch detail, it shall refer to ConductingEquipment (TopologicalNode, branches, etc.). This results in a constraint on interoperability between planning and operation processes.	Requirement
EXCH7.	If a contingency list is exchanged belonging to the model exchanged in node-breaker detail, it shall refer to ConductingEquipment (ConnectivityNode, which is not artificial, Busbar, etc.).	Requirement
EXCH8.	If a model has mixed representation (node-breaker and bus-branch) then the profile URI in the header related to the Equipment Operation is not included as only part of the network will include classes stereotyped with Operation.	Requirement

281

282 5 Specifications and functionalities

283 5.1 General constraints

284 The following rules are general in nature or involve multiple classes. Additional rules are defined in the notes
285 to the individual classes in the profiles part of the CGMES.

ID	Specification	Type
GENC1.	All objects must have a persistent and globally unique identifier (it is the mRID - see Section 5.2). In the ENTSO-E data exchange process this unique identifier will be exchanged as rdf:ID.	Mandatory
GENC2.	Software solutions shall not use “name” related attributes (name, short name, description, etc. inherited by many classes from the abstract class IdentifiedObject) to link the power system model. Only mRID (exchanged as rdf:ID) is used for this purpose.	Mandatory
GENC3.	The rdf:ID defined within a data exchange process is the only globally unique and persistent identifier	Mandatory
GENC4.	IEC 61970-552 defines the rdf:ID as UUID and its syntax (i.e. lower case and number of characters for the different groups part of the UUID). UUID algorithm ensures global uniqueness of the identifier. Example UUID: f81d4fae-7dec-11d0-a765-00a0c91e6bf6	Mandatory
GENC5.	The CGMES defines the identifier as a case sensitive string which conforms to W3C (ISO 8859/1 8-bit single-byte coded graphic character set known as Latin Alphabet No. 1; http://www.w3.org/MarkUp/html3/specialchars.html) with a maximum character limit of 60 characters. A prefix could be added, if necessary, to ensure global uniqueness. The rdf:ID is the mRID plus an underscore “_” added in the beginning of the string.	Mandatory
GENC6.	Applications which conform to the CGMES shall support IEC 61970-552 and rdf:ID expressed as a string.	Mandatory
GENC7.	Each TSO is responsible for ensuring that the rdf:ID is globally unique. ENTSO-E role in ensuring global uniqueness of the rdf:ID is limited to coordination and harmonisation of the approaches used in different data exchanges and which shall conform to GENC4 and GENC5.	Mandatory
GENC8.	rdf:IDs must be kept persistent for all profiles except for State Variable and Diagram layout profiles: • For the State Variable profile the rdf:IDs for state variable classes (SvPowerFlow, SvVoltage, etc.) are not kept persistent. • For the Diagram layout profile the rdf:IDs of DiagramObjectPoint and DiagramObject classes may not be kept persistent	Mandatory
GENC9.	rdf:about expression is used for objects which are exchanged in an instance file for a given profile but defined in a different profile (i.e. exchanged in a different instance file). A stereotype “Description” is introduced to facilitate the implementation of this rule. All classes which shall be expressed by rdf:about are stereotyped with “Description”.	Mandatory

ID	Specification	Type
GENC10.	UTF-8 is the standard for file encoding. UTF-16 is not supported.	Mandatory
GENC11.	Instance data to be exchanged must make use of the most detailed class possible within a profile, i.e. using sub-typed classes rather than general classes e.g. NuclearGeneratingUnit instead of GeneratingUnit.	Mandatory
GENC12.	Optional and required attributes and associations must be imported and exported if they are in the model file prior to import.	Mandatory
GENC13.	If an optional attribute does not exist in the imported file, it does not have to be exported in case exactly the same data set is exported, i.e. the tool is not obliged to automatically provide this attribute. This is not valid if the user is able to process the data, update the model and perform another export.	Mandatory
GENC14.	In most of the profiles the selection of optional and required attributes is made on this basis so as to ensure a minimum set of required attributes without which the exchange does not fulfil its basic purpose. Business processes governing different exchanges can require mandatory exchange of certain optional attributes or associations. Optional and required attributes and associations must therefore be supported by applications which claim conformance with certain functionalities of the CGMES. This provides flexibility for the business processes to adapt to different business requirements and base the exchanges on CGMES compliant applications.	Mandatory
GENC15.	Breakers represent busbar couplers in a bus-branch model exchange. In this case, breakers are only included if they are to be retained. In case of a node-breaker model exchange the rules defined in the IEC 61970-452 and in the CGMES profiles shall be applied.	Mandatory
GENC16.	Roles and multiplicity: The direction of the associations in the profiles part of the CGMES is defined in the profiles. All associations are bidirectional, although an association instance is specified only at one end in the instance files. The documentation of the profiles, which is part of the CGMES, describes the association with the end user. It is allowed to include both ends of an association in the XML, although only the end designated by the profile is required. The following two examples present two options which can be seen in the CGMES profiles: • Example 1: The names “ConductingEquipment.Terminals” and “Terminal.ConductingEquipment” specify opposite ends of the association between the ConductingEquipment class and the Terminal class. In a one-to-many association, the association reference is included with the data of the “many side” class. Therefore, a ConductingEquipment can be associated with up to two Terminals, although a Terminal must be associated with one and only one ConductingEquipment. Consequently, the XML element corresponding to the ConductingEquipment class is not expected to contain any “ConductingEquipment.Terminals” elements. However, the XML element corresponding to the Terminal class is required to contain appropriate “Terminal.ConductingEquipment” elements. • Example 2: The names “TopologicalIsland.TopologicalNodes” and “TopologicalNode.TopologicalIsland” specify opposite ends of the association between the TopologicalIsland class and the TopologicalNode	Mandatory

ID	Specification	Type
	class. The XML element corresponding to the TopologicalNode class is not required to contain any “TopologicalNode.TopologicalIsland” elements. However, the XML element corresponding to the TopologicalIsland class is expected to contain appropriate “TopologicalIsland.TopologicalNodes” elements.	

286

287 ## 5.2 Model authority sets (MAS)

288 The CIM concept of Model Authority Sets is applied to enable the assembly or extraction of TSO models.
289 Model Authority Sets allow an interconnection model to be divided into disjointed sets of objects, which in turn
290 allows different parties to take responsibility for different parts of a common grid model.

ID	Specification	Type
MAS_1.	In any model exchange governed by the CGMES, each model object has an mRID.	Mandatory
MAS_2.	Across all models, the model object instance which represents a given real world asset (line, transformer, etc.) shall always have the same mRID.	Mandatory
MAS_3.	A CIM functional representation is given by CIM classes defined in CGMES UML. Within any one model, object mRIDs are unique, since the same element shall not be represented twice.	Mandatory
MAS_4.	The mRID shall be persistent for the same given functional representation inside a given version of CIM.	Mandatory
MAS_5.	In the case of upgrading from one version of CIM to another the mRID shall be kept persistent for the same functional representation in the new CIM given by a new CIM class. For instance, the change from CIM 15 to CIM 16 allows for a functional representation identified by the mRID for a ShuntCompensator which is changed to the newly introduced class NonLinearShuntCompensator with the same mRID.	Mandatory
MAS_6.	mRIDs are only generated for concrete classes in a given profile. In most cases there is no overlap of functionality between different concrete classes.	Mandatory
MAS_7.	A new mRID is generated in case there is a need to change the class (e.g. GeneratingUnit is changed to ThermalGeneratingUnit). If a physical unit given by mRID in the asset part of the CIM needs to be represented simultaneously as GeneratingUnit and ThermalGeneratingUnit (a given specialisation) it must have different mRIDs for GeneratingUnit and ThermalGeneratingUnit. If different business processes are required to support both types (GeneratingUnit and ThermalGeneratingUnit) the applications should maintain two mRIDs and in both cases support difference file exchange.	Mandatory
MAS_8.	Only one representation (the main class or its specialisation) should be present in a given instance file.	Mandatory
MAS_9.	Each object instance is assigned to one and only one Model Authority Set. There are two types of Model Authority Sets: • Boundary Sets	Mandatory

	• TSO sets	
MAS_10.	Boundary Sets which contain Boundary points marking the boundary between individual models. Boundary Sets are managed by one authority (ENTSO-E) but have been defined by TSOs as a result of mutual agreement.	Mandatory
MAS_11.	TSO sets contain individual TSO instance files. Objects in the TSO sets have internal associations and have associations to Boundary sets. A TSO set shall never have associations with objects in other TSO sets. This allows TSO modelling to be carried out independently of other TSOs	Mandatory
MAS_12.	Each TSO in ENTSO-E is a Model Authority and manages a Model Authority Set in its area of responsibility. The TSO as Model Authority is also responsible for assigning and maintaining object mRIDs in its area set.	Mandatory
MAS_13.	The territory that a TSO model represents may not be exactly the same as the territory managed by a TSO. In the CGMES, “TSO territory” always refers to the model responsibility territory.	Mandatory

5.3 File header

The following rules apply to all headers

ID	Specification	Type
HGEN1.	The definition of file header is specified in IEC 61970-552. The CGMES applies the same definition in any model exchange governed by the CGMES, each model object has an mRID.	Mandatory
HGEN2.	Each type of instance file (full and difference) shall have a file header.	Mandatory
HGEN3.	The file header is declared at the top of the instance file i.e. at the beginning of the file.	Mandatory

The following rules are applied to the model ID (rdf:about) in the file header:

ID	Specification	Type
HREF1.	New ID is generated for new instance files only when the context of instance data changes. An export done on the imported instance data without any changes should have the same model ID reference in the header.	Requirement
HREF2.	Dependent IDs refer to IDs of the dependent instance files at the time of the export.	Requirement
HREF3.	If all dependencies are resolved then there shall not be any unresolved references within the data.	Requirement
HREF4.	The dependency reference in the header shall be used as guidance and shall not restrict the possibility of importing profiles which are exported based on a previous version of a depending profile instance file. The standard does not prevent the tools from exchanging files where the file reference does not match. Unresolved or missing references shall be reported to the user. In general, users are free to combine files on an ad-hoc basis and tooling shall	Requirement

ID	Specification	Type
	identify and optionally resolve all unresolved references.	
HREF5.	Model ID shall be the same if a re-export of a model contains the same objects and attributes. • If the information exchanged with the instance file is the same then the re-export is considered identical. • Rearrangements of classes and attributes in the instance file are allowed.	Requirement

296

297 ## 5.4 File body

ID	Specification	Type
FBOD1.	The IEC 61970-552 specification is used to format a file, although the instance file shall contain only the objects from one Model Authority Set.	Requirement
FBOD2.	An instance file could contain instance data of more than one profile from the CGMES only if all profile URIs are defined in the file header and as long as the instance data belongs to one MAS. All profile URIs are defined in the file header even if one profile is a superset of another, i.e. both URIs should be included.	Requirement
FBOD3.	Instance files may contain objects with associations to objects which will be packaged in a different instance file. This situation means that the instance file by itself is 'incomplete' – it may have dangling references and cannot be used except when combined with one or more other instance file as specified in the file header dependencies. When this occurs, validation for completeness can only be performed when all the parts are present.	Requirement
FBOD4.	The CGMES requires that at the receiving end of the exchange all references in the instance files pointing to instance files from other profiles which are part of the exchange should be satisfied. Therefore, the complete set of instance files necessary for the grid model must have fulfilled references (no dangling references are allowed).	Requirement
FBOD5.	It shall be prevented to create a difference model in case of very small numeric differences as these types of difference models are meaningless.	Requirement

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299 ## 5.5 Profiles and instance file types

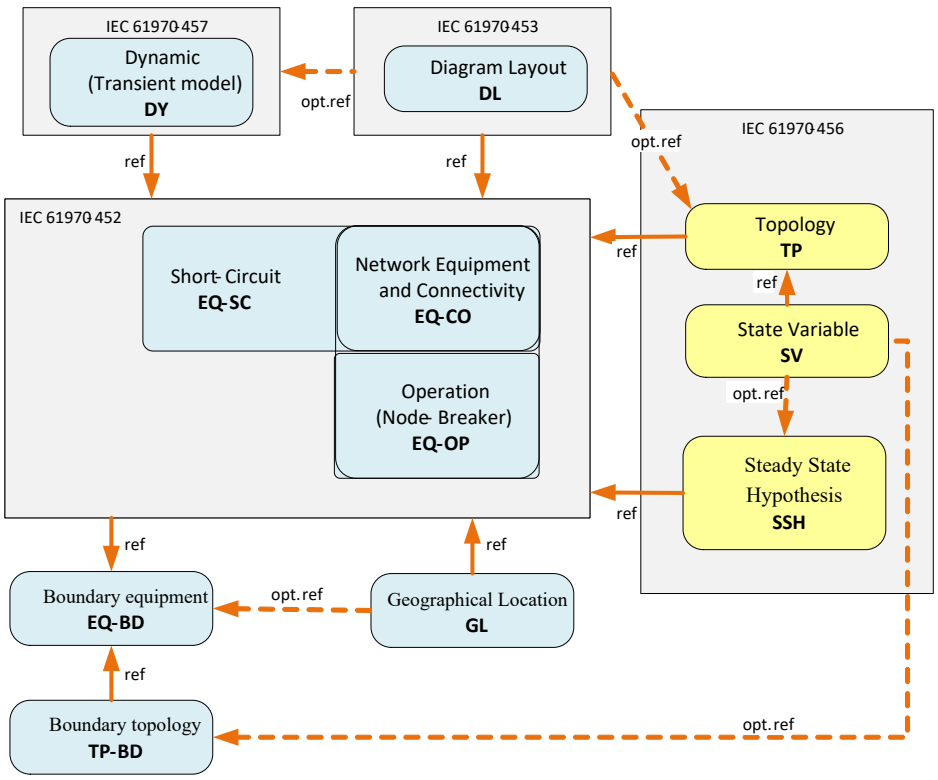
300 There are nine different profiles in the CGMES. This section defines some specific rules for the profiles and
301 their instance files so that the model exchange can be performed correctly.

302 ### 5.5.1 CGMES profiles' properties

303 The profiles which are part of the CGMES are based on IEC CIM UML and maintained in an UML
304 environment.

ID	Specification	Type

ID	Specification	Type
PROF1.	The UML namespace, namespaces of the profiles, ENTSO-E extensions, profiles versions as well as the identification of the versions of the UML and profiles are defined in a Version class for each profile of the CGMES. These properties shall be used as a primary source for file header information.	Requirement
PROF2.	Profile specific notes have been added to various classes and attributes in the UML in order to further clarify different profiles and define specific rules. These notes are considered mandatory and shall be satisfied by the applications	Requirement
PROF3.	Only instances of concrete classes are used in actual exchanges (instance files). Those concrete classes may inherit attributes or associations from abstract classes.	Requirement
PROF4.	The CGMES uses UML stereotypes to categorise classes, attributes and associations used for different exchanges. This is mainly valid for the EQ (equipment) profile where the following categorisation is applied and defines the three different types of equipment instance files supported by CGMES: • EQ core: includes all classes/attributes/associations which are not stereotyped. These elements are part of both node-breaker and bus-branch types of model exchange. • EQ operation: includes all classes/attributes/associations stereotyped with "Operation". These elements are only necessary if a node-breaker model representation is exchanged. • EQ short circuit: includes all classes/attributes/associations stereotyped with "ShortCircuit". These elements are only necessary if a node-breaker or a bus-branch model representation is exchanged for the purpose of performing short circuit calculations	Requirement
PROF5.	The cardinality of given classes/attributes/associations stereotyped with "Operation" or "ShortCircuit" shall be respected if the exchange requires the inclusion of "Operation" or "ShortCircuit". The respective profile URI shall be declared in the file header. If the profile URI is not included in the header all classes/attributes/associations part of the undeclared profile are considered optional.	Requirement
PROF6.	Classes/attributes/associations which were introduced by ENTSO-E and therefore considered as CIM extensions are marked with a stereotype "Entsoe".	Requirement
PROF7.	Any classes/attributes/associations which are defined by the CGMES profiles can be used in grid model exchanges. The authority governing a given business process and related data exchange process shall specify all required attributes/classes/associations to be exchanged. An optional attribute can be required, while a required attribute as defined in the CGMES cannot be changed to optional without modification of the version of the profile. Applications and tools should be able to deal with this complexity and support all classes and attributes depending on the tools' functionalities, i.e. they should at least able to host the data and transfer with no change in case the tool is not able to use the data.	Requirement
PROF8.	All profiles which are part of the CGMES are documented as follows: • UML (XMI), which contains packages “CommonGridModelExchangeStandard” and “Extension”. These packages	Information

ID	Specification	Type
	contain all definitions related to the profiles in the CGMES and are a source of other ways in which to document the CGMES. • RDFS, which contains RDF schema files for each profile of the CGMES. The files are generated by the application CimConteXtor. • HTML, which contains HTML documentation for each profile of the CGMES. The files are generated by the application CimConteXtor. • HTML_EA, which contains the HTML native Enterprise Architect export of the "CommonGridModelExchangeStandard" package. • Pdf, which contains detail description of the CGMES profiles.	
PROF9.	Annex G provides references to the CGMES documentation related to a specific version of the profiles.	Information
PROF10.	The dependencies between the profiles belonging to CGMES are defined in the following figure  The diagram defines the "ref" relations that show a dependency between two profiles. Profile schemas (RDFS) shall not be merged to create a new joint merged profile definitions. Profile instance files shall list in the file header all profiles that describe the instance data so that an application can load all the profile schemas needed to validate the instance data. The "ref" relation may be added to the profile schemas such that an application working with instance data can use this information to load all needed profile schemas. The "ref" and "opt.ref" (optional reference) are used to indicate possible dependencies at instance data level. These dependencies are provided in the file	Requirement

ID	Specification	Type
	header. The blue profiles are invariant, whereas the yellow ones are variant (meaning that the instance files may be different for every other scenario).	
PROF11.	The profile references in the file header specifies for which profiles validation the instance file data is valid for. The instance data file can contain data from multiple profiles (such as ShortCircuit or Operation) without being declared in the header profile references. However, the data belonging to non-declared profiles does not need to be imported and re-exported as the profiles are not defined in the file header. The user shall be informed if the data is not imported.	Requirement

305

306 **5.5.2 CGMES' extensions**

307 Due the complexity and specificity of the ENTSO-E's and TSOs's business processes, the CIM version used
308 to create the profiles of the CGMES has been extended.

ID	Specification	Type
EEXT1.	ENTSO-E extensions defined in the frame of the CGMES and part of its profiles shall be equally supported in the same way as the IEC CIM.	Requirement
EEXT2.	All extension shall be an addition to an existing standard (CGMES of IEC CIM standards). ENTSO-E extensions which are part of the CGMES are extensions of the IEC CIM. When extending the CGMES it is permitted to create a restriction which does not prevent the validity of the CGMES. While an optional attribute or association can be made required as a part of the profiling work, a required attribute or association cannot be made optional as part of an extension.	Requirement
EEXT3.	Declaring the ENTSO-E Extension URI and the Corresponding Alias: The ENTSO-E extension URI and the corresponding alias shall be declared at the topmost element of your CIM/XML file along with CIM and other URI. EXAMPLE (URI of the ENTSO-E CIM extension and the alias): *<?xml version="1.0" encoding="utf-8"?>* *<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"* *xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#* *….* ***xmlns:entsoe="http://entsoe.eu/CIM/SchemaExtension/3/1#">*** *….* *</rdf:RDF>*	Requirement
EEXT4.	Using Extension URI Alias to Declare the Extended Data is illustrated in the following examples. In the first example the attribute IdentifiedObject.shortName is declared as an extended attribute and prefixed with the extension alias "entsoe". EXAMPLE (declaring an extended attribute): *<cim:ACLineSegment rdf:ID="_ f732688e-bace-4ece-bc2b-5d9792608092">* *<cim:IdentifiedObject.name>DFG-THY 1</cim:IdentifiedObject.name>* ***<entsoe:IdentifiedObject.shortName>Brussels</entsoe:IdentifiedObject.s***	Requirement

ID	Specification	Type
	hortName> *<cim:ACLineSegment.bch>4.61157E-05</cim:ACLineSegment.bch>* *<cim:ACLineSegment.r>1.06601</cim:ACLineSegment.r>* *<cim:ACLineSegment.x>34.28535</cim:ACLineSegment.x>* *<cim:Conductor.length>45</cim:Conductor.length>* *<cim:Equipment.aggregate>false</cim:Equipment.aggregate>* *<cim:ConductingEquipment.BaseVoltage rdf:resource=* *"#_ 4852044a-e072-441b-bc3c-dc7b00de7e5e" />* *</cim:ACLineSegment>* In the second example, if an instance of the extended class is declared as extension, then the extended class, i.e. ExtendedClass, and the extended attribute ExtendedClass.extendedAttribute must be prefixed with the extension alias “entsoe”. EXAMPLE (declaring an extended instance or object): ***<entsoe:ExtendedClass rdf:ID="_ 21d3bbfb-0aec-4e44-8db0-ae2e064b22e2">*** *<cim:IdentifiedObject.name>EX11</cim:IdentifiedObject.name>* ***<entsoe:ExtendedClass.extendedAttribute>20</entsoe:ExtendedClass.extendedAttribute >*** *….* *</entsoe:ExtendedClass >* The third example illustrates the addition of an association, namely ENTSO-E extension. EXAMPLE (declaring an extended association): *<cim:ACLineSegment rdf:ID="_9cdc68b4-3953-c88a-24de-820873445a59">* *<cim:ACLineSegment.bch>0.000157</cim:ACLineSegment.bch>* *<cim:ACLineSegment.gch>0</cim:ACLineSegment.gch>* *<cim:ACLineSegment.r>1.04001</cim:ACLineSegment.r>* *<cim:ACLineSegment.x>12</cim:ACLineSegment.x>* ***<entsoe:ConductingEquipment.ExtendedClass rdf:resource="#_9cf549dc-2453-d994-976a-c2ea44773145" />*** *<cim:Conductor.length>30</cim:Conductor.length>* *<cim:IdentifiedObject.name>B1X3</cim:IdentifiedObject.name>* *</cim:ACLineSegment>*	
EEXT5.	The same principle related to extensions applies to any other extensions. Therefore, if an instance file produced by a given application/software contains some extensions specific to vendors' internal applications, the same method to declare such extended data shall be used.	Requirement
EEXT6.	An instant file which contains classes, associations and attributes not defined in the CGMES shall be processed by the receiving application which would ignore the extensions left undefined by the CGMES and make use of the rest of the data.	Requirement
EEXT7.	Extending an enumerator has to be done by adding a new enumerator which includes the additional values. Both the extended and the standard enumerator have to comply with the profile, i.e. if the standard enumerator is mandatory, it shall be included in addition to the new (extended) enumerator.	Requirement

309

5.5.3 Equipment profile and instance file

ID	Specification	Type
EQ__1.	The equipment profile is separated by three functional parts: EQ core, EQ operation and EQ short circuit. The following types of equipment instance files and their relation with the EQ profiles can be exchanged: • Full EQ: contains all classes/attributes/associations defined in EQ core, EQ operation and EQ short circuit profiles. • EQ operation: contains all classes/attributes/associations defined in EQ core and EQ operation profiles. • EQ short circuit: contains all classes/attributes/associations defined in EQ core and EQ short circuit profiles. It covers a bus-branch model exchange which contains short circuit data.	Requirement
EQ__2.	An equipment instance file describes the equipment in the power system model covered by a MAS.	Requirement
EQ__3.	An equipment instance file would not normally change in case of frequent data exchange process. It can be updated with difference file exchange.	Requirement

5.5.4 Topology profile and instance file

ID	Specification	Type
TP__1.	A topology instance file contains all topology objects for a MAS. These topology objects reference the corresponding equipment describing how equipment is electrically connected.	Requirement
TP__2.	A topology instance file is the result of a network topology processing analysis. Because of this the topology instance file is considered as an output if the exchange is based on a node-breaker model exchange.	Information
TP__3.	Depending on the data exchange process a topology instance file may or may not change frequently. E.g. a topology instance file can be updated using difference file exchange in case of bus-branch model exchange.	Information

5.5.5 Steady state hypothesis profile and instance file

ID	Specification	Type
SSH_1.	A steady state hypothesis instance file contains all objects required to exchange input parameters to be able to perform load flow simulations.	Requirement
SSH_2.	A steady state hypothesis instance file is always exchanged in full. Due to the nature of the SSH profile, all objects in an steady state hypothesis instance file shall have persistent mRIDs and rdf:IDs.	Requirement

316 **5.5.6 State variables profile and instance file**

ID	Specification	Type
SV__1.	A state variable instance file contains all objects required to complete the specification of a steady-state solution.	Requirement
SV__2.	A state variables instance file is always exchanged in full.	Requirement
SV__3.	A state variables instance file of an assembled model contains state variables related objects for all model authority sets being part of the assembled model.	Requirement

317

318 **5.5.7 Boundary equipment profile and instance file**

ID	Specification	Type
EQBD1.	A boundary equipment instance file contains all objects defined in the boundary equipment profile and includes data for boundary information relating to a given exchange.	Requirement
EQBD2.	The boundary equipment profile defines which instance data represents types or voltages which are agreed for the CGMES based exchanges. Therefore, individual grid models shall refer to the boundary equipment instance file to use declared EnergySchedulingType-s and BaseVoltage-s. This does not limit different model authorities when it comes to defining additional types or voltages in their instance files, although there shall not be an overlap of data values between boundary equipment files and individual grid model instance files. For instance, ENTSO-E boundary equipment file defines base voltages for both 380 kV and 400 kV to which TSOs instance files should refer and not redefine these base voltages in their instance files.	Requirement
EQBD3.	Boundary equipment instance files can be updated using difference file exchange.	Requirement

319

320 **5.5.8 Boundary topology profile and instance file**

ID	Specification	Type
TPBD1.	A boundary topology instance file contains all objects defined in the boundary topology profile and includes data for boundary information relating to a given exchange.	Requirement
TPBD2.	A boundary topology instance file can be updated using difference file exchange.	Requirement

321

322 **5.5.9 Diagram layout profile and instance file**

323 A diagram instance file is based on the IEC 61970-453 Diagram layout profile standard and contains data
324 necessary for the model diagram.

ID	Specification	Type

ID	Specification	Type
DL__1.	A full Diagram (non-difference instance file) represents a new drawing of the diagram. Data may change from one system drawing to another, e.g. two diagrams with the same mRID of the classes in the instance files do not need to be identical. The purpose of Diagram layout profile is to support the understanding of the equipment data. If a diagram generated by one system is updated by another the file does not need to be identical, with the exception of the edited changes. However, the updated and exported diagram instance file must include all the same relevant information and must have the same layout rendering in the new destination system (old source) as the original, with the exception of the changes. The expected behaviour is that a diagram may have a new layout with the same Diagram mRID as well as DiagramObject mRID. Persistence of Diagram and DiagramObject mRIDs is required if difference updates are supported.	Requirement
DL__2.	The objects in the equipment of the Diagram Layout instance are identified by the DiagramObject.IdentifiedObject.	Requirement
DL__3.	The diagram layout profile supports exchange of more than one diagram, e.g. detailed substation diagram, single line diagram, etc in one instance file of DL profile.	Requirement
DL__4.	A diagram layout instance file is always exchanged in full.	Requirement

5.5.10 Geographical location profile and instance file

ID	Specification	Type
GL__1.	A geographical data instance file contains GIS data and is constructed based on IEC 61968-4, although it is limited to the classes which cover ENTSO-E needs.	Information
GL__2.	A geographical data instance file is exchanged in full, although it could be updated using difference file exchange.	Requirement

5.6 File exchange

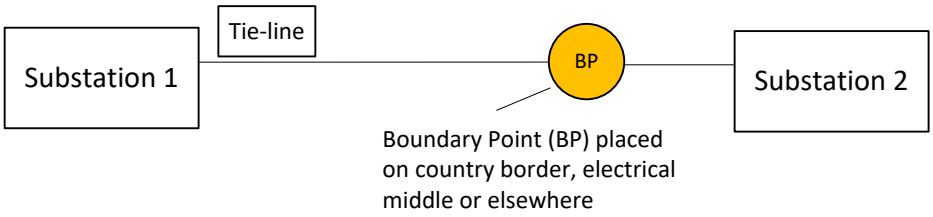
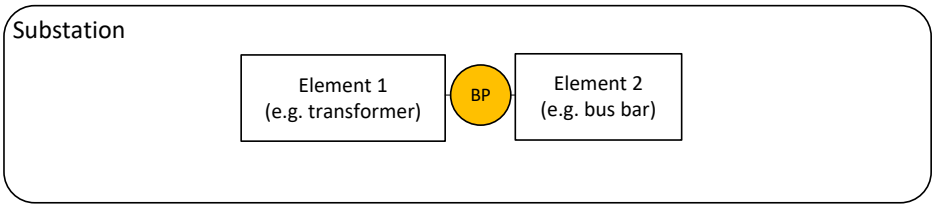
ID	Specification	Type
FILX1.	A given exchange consists of multiple files. The CGMES defines that all files in a given logical exchange must be zipped together[1]. The tools use zip files directly when importing and exporting in order to minimise users.	Information.
FILX2.	The CGMES defines is no naming convention applied to the .xml or .zip file names. Although different business processes may define such a file naming convention, the applications shall rely solely on the information provided in the file headers in order to process the instance files.	Information
FILX3.	One zip file can only contain the following types of files: • A single instance file of the following types: equipment (EQ), boundary	Information

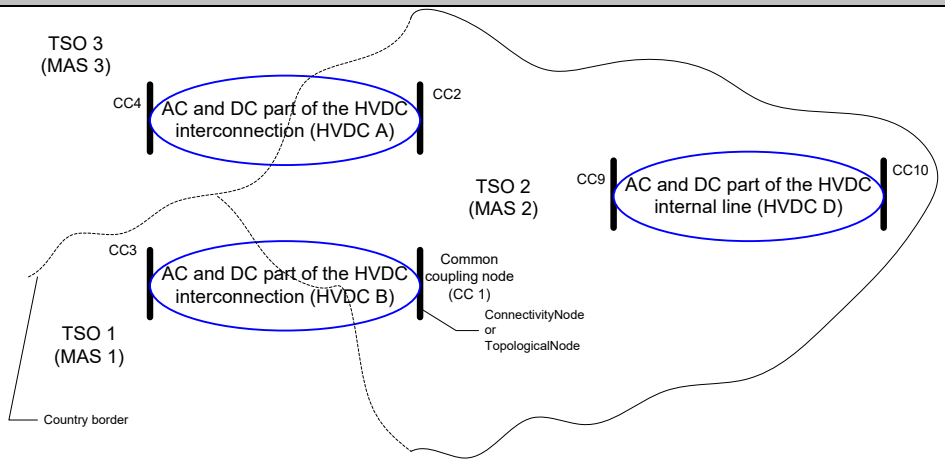
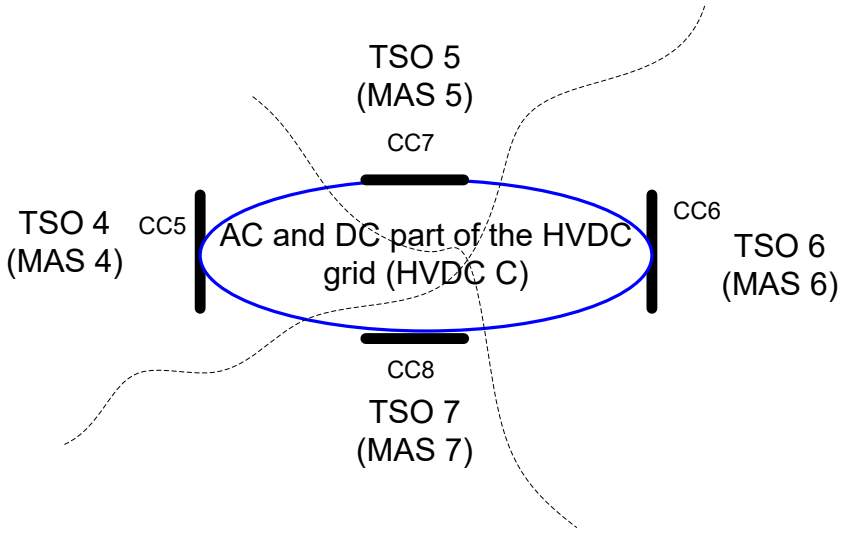
[1] Note that a given business process may require the instances to be exchanged in individual zip files

ID	Specification	Type
	equipment (EQ_BD), topology (TP), boundary topology (TP_BD), steady state hypothesis (SSH), state variables (SV), dynamics (DY), diagram layout (DL), geographical location (GL). • Combinations of equipment, topology, steady state hypothesis, state variables, dynamics, diagram and geographical instance files which are allowed by the CGMES and are related to one MAS only. • Difference files of one MAS only when exchanging a TSO model. • Equipment, topology, steady state hypothesis, state variables, dynamics, diagram and geographical files per MAS for an assembled model. • Difference files per MAS for an assembled model. • Boundary MAS instance files (full or difference or the assembled model is expressed with difference files) shall always be included in the zip file containing an assembled model	
FILX4.	The zip file shall not contain folders. It is only a container of *.xml files.	Requirement
FILX5.	The hierarchy and model dependency shall be respected when exchanging models. The number of files and the type of the files (full or difference) depends on the requirements set by the business process. The following examples show some possible situations: • If the equipment file is changed, all files (depending on the requirements of the exchange: equipment, topology, steady state hypothesis, state variables, dynamics, diagram and geographical files) must be sent as part of any exchange. • If only the steady state hypothesis file is changed, only the steady state hypothesis file must be sent as part of any exchange if there is no requirement to exchange the solved power system model. • If only the state variables file is changed, only the state variables file must be sent as part of any exchange. • If only the dynamics file is changed, only the dynamics file must be sent as part of any exchange. • If only the diagram file is changed, only the diagram file must be sent as a part of any exchange. • If only the geographical file is changed, only the geographical file must be sent as a part of any exchange.	Requirement
FILX6.	It is not allowed to exchange a topology file, a steady state hypothesis file, a state variables file, a dynamics file, a diagram file or a geographical file from one model and an equipment file from another model (or from an entity which has changed the equipment file) and attempt to assemble all files into one assembled model.	Requirement
FILX7.	In case difference files are exchanged, the same dependences as for full model exchange are followed. The difference file (e.g. equipment, topology or dynamics) should refer to the base model which is subject to an update. Dependencies are listed in the file header of each file which is exchanged	Requirement

328

329 ## 5.7 Boundary point – properties and location

ID	Specification	Type
BPPL1.	EquivalentInjection classes are used to represent the power flow exchanges through Boundary points. These classes are included in the individual model MAS (TSO MAS) and refer to the Boundary points in the Boundary set. A SvInjection class is not used for this purpose.	Requirement
BPPL2.	In case the use cases require the exchange of multiple SSH, TP, SV, etc. instance files which are dependent on an EQ instance file, this EQ shall always include an instance of EquivalentInjection per Boundary point. Therefore, in a multi TSO exchange a Boundary point will always have two EquivalentInjections per Boundary point contained in different TSO MAS. Rdf:IDs of those EquivalentInjections are kept persistent.	Requirement
BPPL3.	There are two options related to the location of the Boundary point (BP) in a network model representing the AC grid only: • Boundary point placed on a tie-line: The CGMES does not fix the position of the Boundary point on a tie-line. The Boundary point can be placed on a country border, at the electrical middle of the tie-line, or elsewhere based on mutual agreement between the two neighbouring MAS.  • Boundary point placed in a substation: The CGMES allows a Boundary point to be placed in a substation. The two neighbouring MAS shall agree between which two elements in a substation the Boundary point is placed. 	Requirement
BPPL4.	The CGMES supports HVDC modelling for a detailed representation of HVDC interconnections, TSO internal HVDC links and HVDC grid. The figures below illustrate different cases.	Information

ID	Specification	Type
	 Figure 1 - HVDC as interconnection or internal line  **Figure 2 - HVDC grid**	
BPPL5.	There are two main representations/exchanges of an HVDC link which are supported by the CGMES: • Simplified exchange (no exchange of the AC/DC part of the HVDC interconnections. A HVDC link is represented with two radial AC lines) • Detail exchange (AC/DC part of HVDC links is exchanged)	Information
BPPL6.	In the simplified exchange of an HVDC link the net interchange between the MAS is represented by EquivalentInjection classes referring to each common coupling node (CC).	Requirement
BPPL7.	The simplified exchange of an HVDC link could be applied to internal HVDC links (systems) as well as to HVDC interconnections.	Information

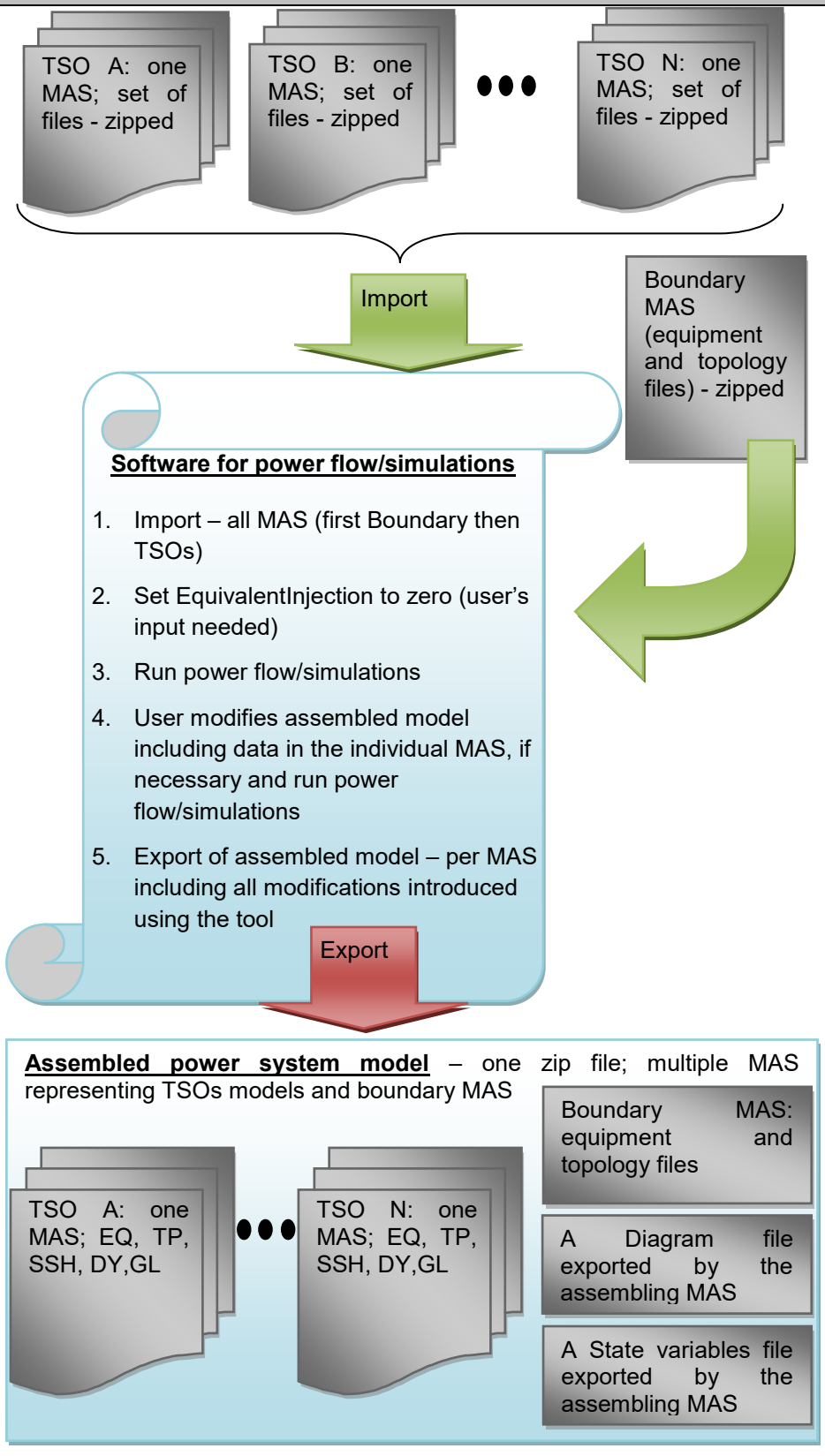
ID	Specification	Type
BPPL8.	In the detail exchange of an HVDC link the HVDC grid shall be exchanged as a MAS: • Separate instance files (EQ, TP, SSH, SV) are included in this MAS. • In case one TSO is responsible for the HVDC, the HVDC model is included in the TSO MAS (EQ, TP, SSH, SV).	Requirement
BPPL9.	In case of a detail exchange of an HVDC link, the HVDC MAS shall refer to the common coupling points (ConnectivityNode or to TopologicalNode) included in the Boundary set.	Requirement
BPPL10.	In case of more than one HVDC interconnection (including interconnections with different TSOs) the TSO can include them in a single HVDC MAS or in its own MAS.	Information
BPPL11.	In particular cases a Boundary point can be placed on a DCLineSegment and could possibly represent a different authority. The CGMES does not allow for the separation of the HVDC model at this specific Boundary point. It only makes sense to add a Boundary point with a location to identify certain responsibilities	Information

330

331 ## 5.8 Model assembling process

ID	Specification	Type
MAPR1.	A complete (assembled) common grid model (solved or unsolved power system model) contains information from more than one model authority set. Part of the reason for the division into files per MAS is to create better flexibility when it comes to how complete assembled models for different purposes are formed from base parts. Model management systems can be designed based on this capability.	Information
MAPR2.	Among instance files which are to be combined to form an assembled model, there is no overlap – each object, association or attribute appears in one and only one of the instance files being combined.	Requirement
MAPR3.	The model assembling procedure in the CGMES is based on the Model Authority Sets concept. The procedure includes the following steps: • Model management system (calculation tool/software) imports all MAS (full set of files for MAS for each TSO and Boundary MAS). Depending on the implementation of the import process, the Boundary MAS must be imported first in case other MAS are subsequently imported. The following files/MAS should be available for import. o At least two models from TSOs are available and represented in two different MAS. These models have necessary references to the Boundary set. o TSO models which include classes (EquivalentInjection for SSH profile; SvPowerFlow and SvVoltage for SV) represent the flow between the MAS and the voltage of the Boundary points. o Boundary MAS: Boundary instance files (equipment and topology) cover, but are not limited to, the area represented in the common grid	Requirement

ID	Specification	Type
	model. • SvVoltage classes pointing to the Boundary set are set to the nominal voltage of the Boundary Node with zero angles in case the values for EquivalentInjection pointing to a Boundary Node are different. • EquivalentInjection.p and EquivalentInjection.q are set to zero, if a tie-line or a ConnectivityNode is connected. It is not necessary that these parameters be set to zero at the time of the import. Additional functions should be made available for users to cover all necessary use cases when dealing with TSO MAS and Boundary MAS. • A power flow can be performed to obtain a solution for the assembled power system model.	
MAPR4.	An update of the assembled power system model is performed via an update of the concerned MAS (i.e. replacing of MAS files). A power flow solution is necessary to update the common state variables file valid for the updated assembled model.	Requirement
MAPR5.	State variable instance file for the common grid model include SvPowerFlow for injections of Boundary points. In order to solve an assembled model these injections are set to zero in case a Boundary point successfully connects the two MAS. Some injections (SvPowerFlow) may differ from zero to represent the exchange with other areas not included in the assembled model.	Requirement
MAPR6.	The exported assembled model shall have only one instance of SvVoltage per Boundary point.	Requirement
MAPR7.	The CGMES does not fix the content of an assembled grid model. Different business processes shall define the type (EQ, SSH, TP, SV, etc.) of data needed in the assembled model depending on the objectives.	Information
MAPR8.	The CGMES supports export of unsolved assembled model. The model can be a node-breaker or bus-branch model representation type, and shall always include SSH instance files if the purpose of the exchange is to perform (without data additions) a load flow calculation in a different application.	Information
MAPR9.	The exported assembled model contains multiple MAS. These exported multiple MAS include all changes which are introduced on the assembled model (changes made in the software which is used to assemble the model) per MAS. The following chart illustrates the assembly process:	Requirement

ID	Specification	Type
		
MAPR10.	The model authority set of the state variable instance file of the assembled model is not defined in the file header.	Requirement

ID	Specification	Type
MAPR11.	The model authority set of the diagram layout instance file of the assembled model is not defined in the file header.	Requirement

5.9 RDF/XML model validity

In order to be considered a valid model, a given combined set of RDF/XML must adhere to the following criteria:

ID	Specification	Type
MVAL1.	The file must be well-formed as defined by the Extensible Markup Language (XML) 1.0 (Second Edition) (http://www.w3.org/TR/REC-xml).	Requirement
MVAL2.	The file must adhere to the rules set forth in the Simplified RDF Syntax for Power System Model Exchange as defined in IEC 61970-552.	Requirement
MVAL3.	The file must contain CIM entities which are valid according to the CIM RDF Schema file.	Requirement

5.10 Naming Convention

The naming convention of any profile is important in order to ensure that the information which is part of the data exchange can be understood and used. A power system model without appropriate naming information cannot be readily used for any human analyses and thus loses its meaning.

The CGMES, which uses multiple profiles, serves various business processes. These business processes have different needs in terms of naming information. There is no single set of rules which could be applied to deal with the specificities of different business processes axes such as:

- Bilateral, regional, ENTSO-E – pan-European data exchanges;
- Operational (day ahead and related processes), long term planning data exchanges;
- Node-breaker, bus-branch based data exchanges;
- Voluntary, project oriented, obligatory by an ENTSO-E process, obligatory by law data exchanges.

A restriction related to naming could serve one business process well but may represent a significant constraint for another business process. In addition, actors involved in the exchange are not necessarily the same.

Naming is a complex issue as different applications or processes can have naming conventions that are inconsistent or conflicting with naming conventions used by other applications or processes.

CGMES applies the following naming principles:

- Names are mainly for humans. They shall not be used for any purposes object identification.
- Naming rules shall be driven by the use cases (mainly information presentation), not restricted by limitations of vendors systems and UML/profiling tools.
- Names are important in many cases. But restrictive requirements on naming may result in the interoperability issues. For example, the dynamically-created names may not be storable in the underlying data source, resulting in information loss.

- Physical modelling entities, such as Substation, Equipment, etc., require a name to be specified, while naming is optional for the components that make up a physical modelling entity, such as Terminal, RatioTapChanger, etc.
- Conceptual modelling entities may or may not require a name to be specified depending on whether these modelling entities shall be presented to end-users. DiagramObjects, for example, are meaningless for human audiences. TopologicalIsland, on the other hand, may require a name to be provided, since they could be shown in the power flow reports.
- Name of conceptual modelling entities shall be optional whenever possible. For example, if the conceptual modelling can be characterized by one or a combination of its attributes, then naming shall be optional. As an example, name for BaseVoltage may not be required, since it is characterized by attribute BaseVoltage.nominalVoltage, which could optionally serve as name. By the same token, if a conceptual modelling entity is uniquely associated with a physical modelling entity, then its name can be derived from the associated physical modelling entity if needed.

Therefore, the objective of the CGMES naming convention is to define a common framework related to naming rules which could be further restricted by different business processes.

The following rules related to the naming convention are defined:

ID	Specification	Type
NAMC1.	A template for further defining constraints related to the naming convention is provided in Annex A. The template shall be used by experts defining naming convention rules within a business process. The template shall be completed by a profile (EQ, TP, SV, etc.). As soon as a business process is defined and data exchange requirements agreed, the tables related to different profiles should be made available to all parties participating in the data exchange and vendors should be informed. This will allow: • TSO experts to be aware and respect requirements related to naming in the models. • Vendors developing tools for power system analyses to ensure that: o TSO experts are able to supply the names as required by the business process; o An agreement for naming translation between requirements in the business process and the proprietary formats or TSO databases is in place; o Export and import functionalities are compliant with the data exchange rules. • Vendors developing validation tools to adjust validation rules valid for the business process	Information
NAMC2.	The restrictions related to the naming convention are considered obligatory for any tool importing or exporting naming data if the tool claims compliance with the CGMES.	Requirement
NAMC3.	Further restrictions can be applied by different business processes. Business processes restrictions on naming must define required and optional attributes related to naming. These restrictions or rules should not contradict this naming convention and are considered mandatory for all parties participating in a given	Requirement

ID	Specification	Type
	business process. • Tools used to validate instance data shall be able to validate against different sets of naming conventions which are applied to exchanges based on the CGMES. • Tools used for various power system analyses shall provide users with the opportunity to cope with different naming rules. These tools shall be developed to support the full scope of this naming convention.	
NAMC4.	Due to the current inheritance structure of the CIM used for the profile, the naming convention primarily addresses the attributes of the class IdentifiedObject. However, there are certain exceptions, including Boundary profiles whereby some ENTSO-E extensions are applied to ConnectivityNode and TopologicalNode.	Information
NAMC5.	The name related attributes have an informational character intended for human reading for explanations outside the classes. Software solutions must not count on this information to complete physical links of the power system model. All necessary links between different parts of the CIM XML are expressed by the reference schema which uses rdf:ID	Requirement
NAMC6.	Names shall conform to UTF-8	Requirement
NAMC7.	In cases where tools using instance data (compliant with the CGMES exchanges) need uniqueness rules, this shall be handled in the importing function based on requirements defined by the users.	Requirement
NAMC8.	It is obligatory that information exchanged in name related attributes is not modified by the tools within an exchange, i.e. • step 1: Tool A imports data from Tool B and modifies initial information to fit user requirement of tool limitation; • step 2: Tool A is obliged to export the imported data in the same form and content as the data exported from Tool B. One of the main reasons behind this rule is the fact that exchanges in the ENTSO-E are meant to be bi-directional, i.e. there is a sending party and a receiving party which exchange models within studies and do not necessarily consume only the data.	Requirement
NAMC9.	It is mandatory that tools shall provide users with the ability to add and maintain naming related information for classes which represent physical equipment as well as classes which represent elements important for business processes (e.g. TopologicalNode). This information is then mapped onto relevant attributes and can be exported for the purpose of the exchange.	Requirement
NAMC10.	There is no need for a specific naming convention when it comes to the names of the instance files due to file header information which is defined by IEC CIM standards and the CGMES.	Information
NAMC11.	The following tables summarise the use of name related attributes in the different profiles. It provides the length of the strings which tools must support for all classes that inherit from IdentifiedObject or on ConnectivityNode and TopologicalNode.	Requirement

<table>
<tr><th>ID</th><th>Specification</th><th>Type</th></tr>
<tr><td></td><td>

Table 1 — IdentifiedObject attributes

IdentifiedObject	String length, characters	Equipment profile	Topology profile	Study State Hypothesis profile	State variables profile	Diagram layout profile	Geographical location profile	Dynamics profile
.name	32 max	✓r	✓o	✓o	✓r	✓r	×	✓o
.description	256 max	✓o	✓o	×	×	×	×	✓o
.energyIdentCodeEic	16 exactly	✓o	✓o	×	×	×	×	×
.shortName	12 max	✓o	✓o	×	×	×	×	×

Table 2 — IdentifiedObject attributes for ConnectivityNode in EQ_BD profile and for TopologicalNode in TP_BD profile

IdentifiedObject	String length, characters	Boundary equipment profile	Boundary topology profile
.name	32 max	✓r	✓r
.description	256 max	✓r	✓r
.energyIdentCodeEic	16 exactly	✓o	✓o
.shortName	12 max	✓r	✓r
.fromEndIsoCode	2 exactly	✓r	✓r
.toEndIsoCode	2 exactly	✓r	✓r
.fromEndName	32 max	✓r	✓r
.toEndName	32 max	✓r	✓r
.fromEndNameTso	32 max	✓r	✓r
.toEndNameTso	32 max	✓r	✓r

Legend: ✓r – the attribute is present in the profile and required (required means that it is mandatory that this attribute be present in the instance data); ✓o - the attribute is present in the profile and optional; × - the attribute is not present in the profile.

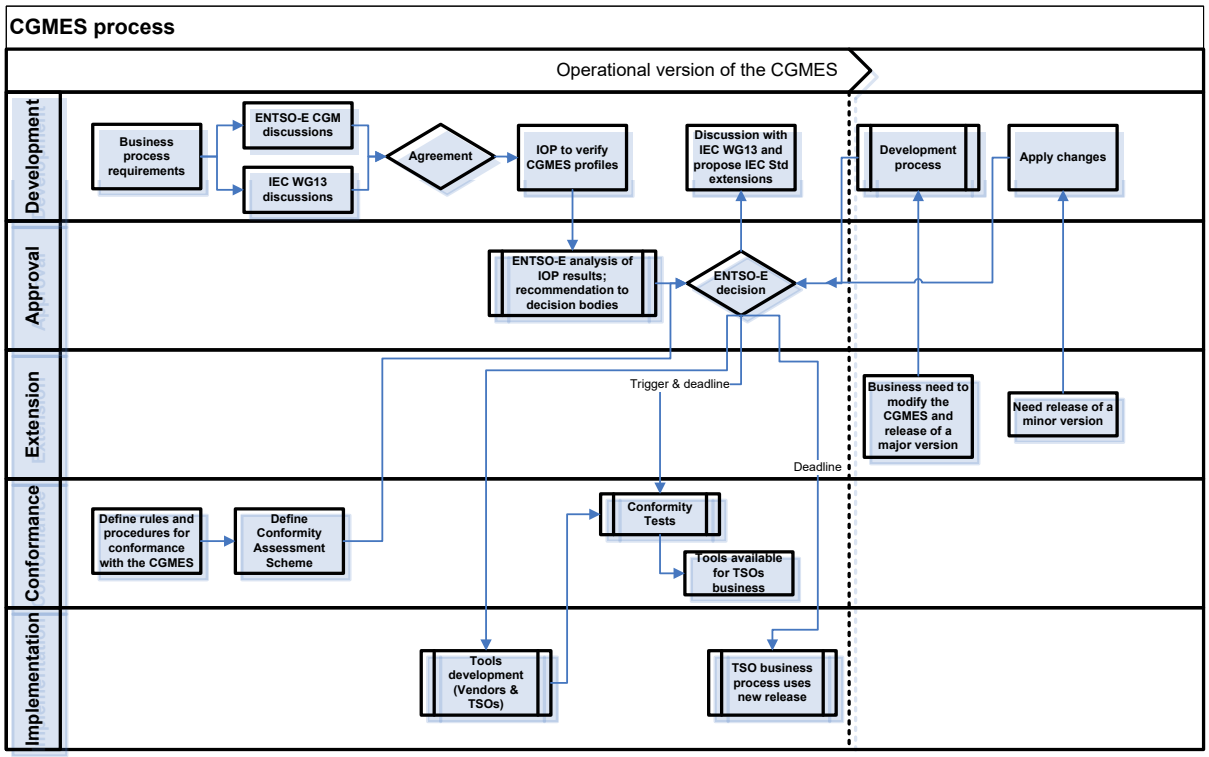
</td><td></td></tr>
<tr><td>NAMC12.</td><td>Additional specific rules related to the naming attributes are included as notes in the profile specifications. These notes are considered as mandatory rules. The same rules are summarised in Annex B.</td><td>Requirement</td></tr>
<tr><td>NAMC13.</td><td>OCL validation rule in the TP profile defined in which cases the attribute IdentifiedObject.name is required.</td><td>Requirement</td></tr>
<tr><td>NAMC14.</td><td>Name related attributes of the IdentifiedObject that are required can be exchanged as empty string i.e. <IdentifiedObject.name></IdentifiedObject.name></td><td></td></tr>
</table>

6 CGMES governance

The overall governing process of the CGMES is a complex process and includes the following sub-processes:

- Standardisation process – a process to develop the CGMES, and which relies on the latest IEC CIM related standards.
- Interoperability process – the process to conduct IOPs targeting verification of the CGMES and IEC CIM standards. Feedback to standardisation bodies and organisations involved in the development of different profiles which form part of the CGMES.
- Business process – all business processes which use the CGMES, They define requirements and request additional improvements due to business needs.
- CGMES conformance process – this process ensures that tools used by parties involved in an exchange utilising the CGMES are implementing the CGMES correctly.
- Implementation process – the implementation process is triggered by a business process/need. It aims to apply a certain version of the CGMES to a business process.

The following chart illustrates the main stages related to the CGMES.



6.1 Versions of the CGMES and the profiles

Each version of the CGMES and the profiles part of the CGMES has its unique version identifier. The following rules related to versioning are defined:

ID	Specification	Type
VERS1.	The format of a version of the CGMES is xx.yy.zzz where xx, yy and zzz are non-negative integers, and must not contain leading zeroes, and: • xx - names the major version of the CGMES • yy – names the minor version of the CGMES	Requirement

ID	Specification	Type
	• zzz –names the revision version of the CGMES	
VERS2.	Each profile part of the CGMES is assigned with a version defined by the profile URI which shall be declared in the file header of the instance files. The profile URI is specified in the UML of the CGMES.	Requirement
VERS3.	A profile URI changes every time a minor or a major version of a profile is released.	Requirement
VERS4.	The namespace URI of the ENTSO-E extensions changes every time a minor or a major version of the extension package is released.	Requirement
VERS5.	The namespace UML changes every time the CGMES changes the base version of the CIM, e.g. the base UML changes from CIM 16 to CIM 17.	Requirement
VERS6.	Each of the CGMES profiles is related to a profile defined by the IEC. The ENTSO-E UML lists the base URI of the IEC profiles for information only and to link a specific profile of the CGMES to the closest IEC CIM profile. This information is provided in the base URI attribute of the version class to each profile of the CGMES	Requirement
VERS7.	A minor version is a compatible change to a profile. The minor version must be incremented if new, backwards compatible functionality is introduced to the CGMES. It must be incremented if any functionality is marked as deprecated. It may be incremented if substantial new functionality or improvements are introduced to the CGMES by adding additional profiles and/or ENTSO-E extension. It may include revision level changes. Revision version must be reset to 0 when minor version is incremented.	Requirement
VERS8.	The number of a major must be incremented if any backwards incompatible changes are introduced to the CGMES, e.g. something is deleted. It may include minor and revision level changes if the amount of changes is significant. The major version will also be incremented if one or more profile is no longer backwards compatible. Minor and revision version numbers must be reset to 0 when major version is incremented.	Requirement
VERS9.	Updates belonging to a minor version update should not break the interoperability for a major profile exchange. This means that a tool which supports a profile version, e.g. 2.4 (2 major version, and 4 minor version) shall be able to import a file which is generated based on profile version 2.5 where all the additional classes, attributes and associations are ignored.	Requirement
VERS10.	The revision version must be incremented if only backwards compatible error fixes are introduced. A fix is defined as an internal change that fixes incorrect behaviour. Updating documentation or a class, an attribute or a profile to reflect the intended behaviour are considered error fixes.	Requirement
VERS11.	A pre-release version may be denoted by appending a hyphen and a series of dot separated identifiers immediately following the revision version. Identifiers must comprise only ASCII alphanumeric and hyphen [0-9a-Za-z-]. Identifiers must not be empty. Numeric identifiers must not include leading zeroes. A pre-release version indicates that the version is unstable and might not satisfy the intended compatibility requirements as denoted by its associated normal version. Example: 2.5.0-alpha.	Requirement

ID	Specification	Type
VERS12.	Once a versioned package has been released, the contents of that version must not be modified. Any modifications must be released as a new version.	Requirement
VERS13.	The key words "must", "must not", "required", "shall", "shall not", "should", "should not", "recommended", "may", and "optional" in this section of the CGMES are to be interpreted as described in RFC 2119.	Requirement

6.2 Conformity assessment

Conformity assessment of tools is necessary to confirm that tools comply with a given profile part of the CGMES and can be used for model exchange in a given business process. Conformity assessment is business driven and ensures reliability of the model exchanges by confirming interoperability between applications. The conformity assessment processes that shall be followed is defined in the ENTSO-E CGMES Conformity Assessment Framework.

Each new version of a tool shall be tested for conformity with a particular version of the CGMES used in ENTSO-E business processes prior to its usage in the business processes. ENTSO-E members are responsible for ensuring that tools which they use in the frame ENTSO-E business processes conform to the CGMES.

Test configurations (models) representing the main functionalities of the profiles of the CGMES shall be publicly available to all interested parties no later than 3 months after the approval of a major or minor release of the CGMES or its profiles. Depending on the complexity of the changes in the profiles the decision body approving the CGMES shall either confirm this deadline or specify another deadline.

Each new version of the CGMES shall include information on which conformity assessment procedures and test configurations should be updated.

Conformity assessment shall rely on a machine readable way of defining the validation rules and describing the constraints valid for a certain profile. Object Constraint Language (OCL) is used for this purpose.

6.3 Implementation process

The implementation of a version of the CGMES for use in a business process is launched as soon as the conformity assessment process has been finalised. The following rules are defined for the implementation process:

The implementation process is triggered by the body responsible for the model exchange. The body defines the deadline when the implementation process shall end and the business process switches to the new version of the CGMES.

The implementation process includes a period during which TSOs shall upgrade their tools and a period during which a trial tests running the business process with the new version of the CGMES. This is especially valid for operational exchanges where the exchange shall be reliable and completed more frequently than a planning model exchange process.

The CGMES contains various profiles. The implementation of each of them can have a different schedule depending on the business needs.

Due to the different requirements of the business processes, different versions of the CGMES profiles can be simultaneously operational. The ENTSO-E Secretariat shall maintain a publicly available list of version of the CGMES used in business processes. Vendors and TSOs shall adapt the implementation process and the support to the model exchange processes and be able to cope with a variety of the CGMES versions used in the exchanges.

435 Business processes shall be adapted in order to allow smooth interfaces between main types of exchange
436 such as planning to planning, operation to operation, operation to planning, interface with distribution, etc.
437 Business processes shall aim to use a limited number of different versions of the CGMES in order to decrease
438 maintenance effort by TSOs and facilitate interoperability of data exchanges between business processes.

Annex A
(normative)

Template for further restrictions on naming

Class name	IdentifiedObject							
	.name		.description		.energyIdentCodeEic		.shortName	
	Status [✓r, ✓o, ×]	Required content/rules	Status [✓r, ✓o, ×]	Required content/rules	Status [✓r, ✓o, ×]	Required content/rules	Status [✓r, ✓o, ×]	Required content/rules

Legend: ✓r – the attribute is required; ✓o - the attribute is optional; × - users do not need to have the ability to add name to the attribute.

Annex B
(normative)

Summary of specific rules for naming convention

B.1 IdentifiedObject.name

Human readable name with a maximum string length of 32 characters. The length restriction is defined in the IEC 61970-452.

IdentifiedObject.name is a required attribute for the Boundary Equipment and Boundary Topology profiles.

The IdentifiedObject.name shall be consistent with the name of the object used in companies, in daily operation (e. g. in SCADA systems), in planning processes or in asset related systems.

The IdentifiedObject.name should allow for inter-communicating of TSO, using general names.

B.2 IdentifiedObject.description

Human readable description with maximum string length of 256 characters. The length restriction is defined in the IEC 61970-452.

IdentifiedObject.description is a required attribute for the Boundary Equipment and Boundary Topology profiles.

B.3 IdentifiedObject.energyIdentCodeEic

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code.

References:

- Local issuing offices for EIC: https://www.entsoe.eu/publications/electronic-data-interchange-edi-library/links-to-eic-websites/
- EIC description: https://www.entsoe.eu/index.php?id=73&libCat=eic

B.4 IdentifiedObject.shortName

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.”

B.5 ConnectivityNode and TopologicalNode .fromEndIsoCode

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of the ISO code of the region to which the “from” side of the Boundary point belongs to or is connected to.

The ISO code is a two character country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum.”

The attribute is required for the Boundary Model Authority Set where this attribute is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.

B.6 ConnectivityNode and TopologicalNode .toEndIsoCode

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of the ISO code of the region to which the “to” side of the Boundary point belongs to or is connected to.

The ISO code is a two character country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum.”

The attribute is a required for the Boundary Model Authority Set where this attribute is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.

B.7 ConnectivityNode and TopologicalNode .fromEndName

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of a human readable name with length of the string 32 characters maximum. The attribute covers two cases:

- if the Boundary point is placed on a tie-line the attribute is used for exchange of the geographical name of the substation to which the “from” side of the tie-line is connected.
- if the Boundary point is placed in a substation the attribute is used for exchange of the name of the element (e.g. PowerTransformer, ACLineSegment, Switch, etc) to which the “from” side of the Boundary point is connected.”

The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.

B.8 ConnectivityNode and TopologicalNode .toEndName

It is an ENTSO-E extension of the CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of a human readable name with length of the string 32 characters maximum. The attribute covers two cases:

- if the Boundary point is placed on a tie-line the attribute is used for exchange of the geographical name of the substation to which the “to” side of the tie-line is connected.
- if the Boundary point is placed in a substation the attribute is used for exchange of the name of the element (e.g. PowerTransformer, ACLineSegment, Switch, etc) to which the “to” side of the Boundary point is connected.”

The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.

B.9 ConnectivityNode and TopologicalNode .fromEndNameTso

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of the name of the TSO to which the “from” side of the Boundary point belongs or is connected to. The length of the string is 32 characters maximum.

The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.”

B.10 ConnectivityNode and TopologicalNode .toEndNameTso

It is an ENTSO-E extension of the IEC CIM. The following description was given in the UML for this attribute: “The attribute is used for an exchange of the name of the TSO to which the “to” side of the Boundary point belongs or it is connected. The length of the string is 32 characters maximum.

The attribute is required for the Boundary Model Authority Set where it is used only for the TopologicalNode in the Boundary Topology profile and ConnectivityNode in the Boundary Equipment profile.”

B.11 Future developments on CIM for dynamics

The following gaps and needs for future development are currently identified. Discussion regarding these topics is expected to:

- Extend user-defined models approach of the dynamics profile.
- The class StaticVarcompensator is present in the equipment model, but is not represented in the dynamic model. The class could probably be used to represent different FACTS components, although it should be possible to define type of SVC (TCR,TCR/TSC, STATCOM) as this is of importance when it comes to how dynamic response is modelled.
- There are no models representing branch quantities (like line relays and models which measure line flow). It is critical that this type of model should be tested at the IOP, since the input and behaviour is very different from unit models (topology dependency).
- Clarification regarding wind farms modelling is necessary. There is a need to verify that the models are compliant with the last IEC standard on wind modelling. There is also the need for adequate modelling of the wind power plants/parks.
- There are no relay models at all, whether for lines or for loads/units. Frequency dependent load relays are important.
- Models to represent dynamics behaviour of HVDC are necessary.

Annex C
(normative)

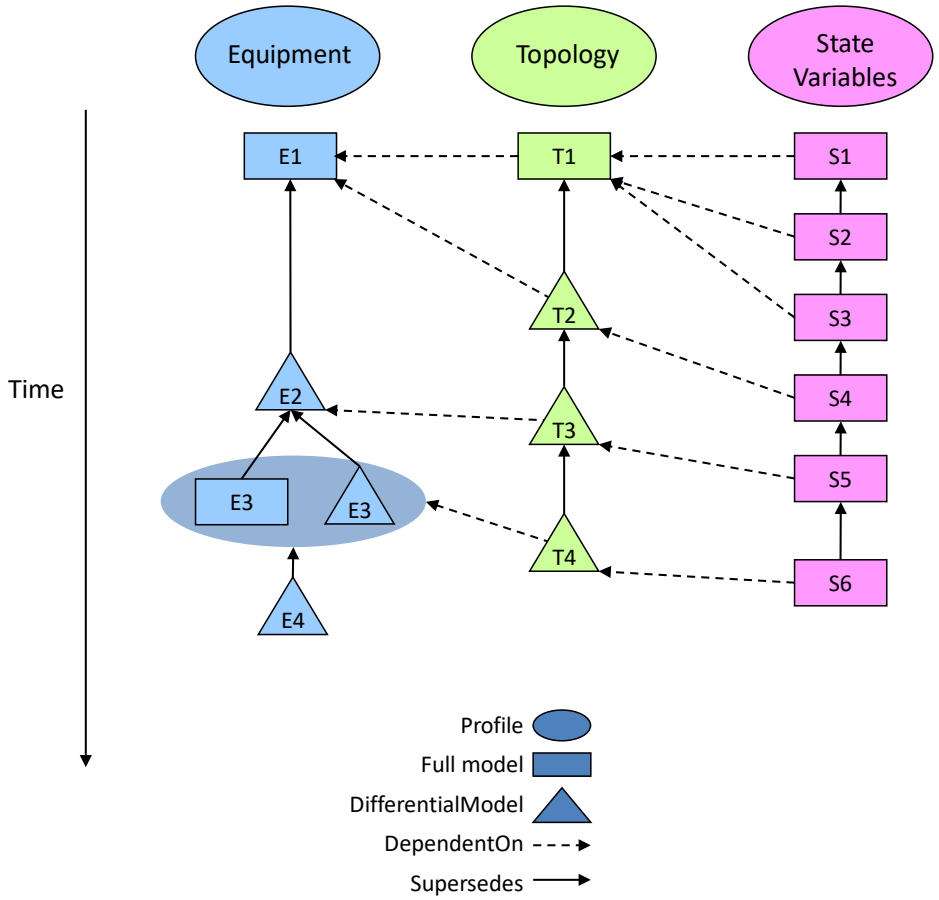
File header guidelines

C.1 Introduction

The annex is providing guidance on how to apply IEC 61970-552 which is defining file headers for complete instance files and difference instance files.

C.2 Exchange scenarios

The IEC 61970-552 gives as an example the following figure which is used in order to explain what information should be provided in different file headers.



C.3 Examples

C.3.1 Example 1: File header of full model

This example is for full model E1 from the figure in section C.2.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	
<md:FullModel rdf:about="urn:uuid:bcb6877a-e948-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>1</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/EquipmentCore/3/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case EQ instance file contains one profile.
<md:Model.profile>http://entsoe.eu/CIM/EquipmentOperation/3/1</md:Model.profile> <md:Model.profile>http://entsoe.eu/CIM/EquipmentShortCircuit/3/1</md:Model.profile>	Additional profile URN/URI are included if classes and attributes that belong to Operation and Short Circuit are used and the instance file is intended for such use.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet> </md:FullModel>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.

C.3.2 Example 2: File header of full model that is depending on another model

This example is for full model T1 from the figure in section C.2.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	
<md:FullModel rdf:about="urn:uuid:d63e4784-e94b-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>1</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/Topology/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the TP profile.
<md:Model.DependentOn rdf:resource="urn:uuid:bcb6877a-e948-11e3-89cf-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the T1 model depends on E1 model. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet> </md:FullModel>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of "frames" which is under discussion.

C.3.3 Example 3: File header of full model that is depending on a model and supersedes another model

This example is for full model S2 from the figure in section C.2.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	
<md:FullModel rdf:about="urn:uuid:aad4c9cc-e94d-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the "urn:uuid:" is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>2</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/StateVariables/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the SV profile.
<md:Model.DependentOn rdf:resource="urn:uuid:d63e4784-e94b-11e3-89cf-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the S2 model depends on T1 model. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.Supersedes rdf:resource="urn:uuid:b1f4d506-e94f-11e3-89cf-82687f4fc15c"/>	When a model is updated the resulting model supersedes the models that were used as basis for the update. Hence this is a reference to the CIMXML documents which are superseded by this model. A model (or instance file) can supersede 1 or more models, e.g. a difference model or a full model supersede multiple models (difference or full). In this case more than one Model.Supersedes are included in the header. The referenced document(s) is (are) identified by the URN/MRID/UUID in the FullModel

	rdf:about attribute when full model(s) is (are) referenced and by the URN/MRID/UUID in the DifferenceModel rdf:about attribute when difference model(s) is (are) referenced. In this case the model S2 supersedes model S1 which has <md:FullModel rdf:about="urn:uuid:b1f4d506-e94f-11e3-89cf-82687f4fc15c">
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet> </md:FullModel>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of “frames” which is under discussion.

C.3.4 Example 4: File header of difference model that is depending on a full model and supersedes another full model

This example is for difference model T2 from the figure in section C.2.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:dm="http://iec.ch/TC57/61970-552/DifferenceModel/1#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	Please pay attention that for difference models xmlns:dm=http://iec.ch/TC57/61970-552/DifferenceModel/1# is included as well
<dm:DifferenceModel rdf:about="urn:uuid:7ca72efa-e952-11e3-89cf-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the “urn:uuid:” is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>2</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/Topology/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and

	its version. In this case it is the profile URI of the TP profile.
<md:Model.DependentOn rdf:resource="urn:uuid:bcb6877a-e948-11e3-89cf-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the T2 model depends on E1 model. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.Supersedes rdf:resource="urn:uuid:d63e4784-e94b-11e3-89cf-82687f4fc15c"/>	When a model is updated the resulting model supersedes the models that were used as basis for the update. Hence this is a reference to the CIMXML documents which are superseded by this model. A model (or instance file) can supersede 1 or more models, e.g. a difference model or a full model supersede multiple models (difference or full). In this case more than one Model.Supersedes are included in the header. The referenced document(s) is (are) identified by the URN/MRID/UUID in the FullModel rdf:about attribute when full model(s) is (are) referenced and by the URN/MRID/UUID in the DifferenceModel rdf:about attribute when difference model(s) is (are) referenced. In this case the model T2 supersedes model T1, i.e. this difference model is applied on the model which has the id mentioned in this Model.Supersedes.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of “frames” which is under discussion.
Here the place for the content of the difference file	This follows the syntax as defined in the 61970-452.
</dm:DifferenceModel>	

C.3.5 Example 5: File header of difference model that is depending on a difference model and supersedes another difference model

This example is for difference model T3 from the figure in section C.2.

Syntax	Comment/Explanation
<?xml version="1.0"?> <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:cim="http://iec.ch/TC57/2013/CIM-schema-cim16#" xmlns:dm="http://iec.ch/TC57/61970-552/DifferenceModel/1#" xmlns:md="http://iec.ch/TC57/61970-552/ModelDescription/1#" xmlns:entsoe=http://entsoe.eu/CIM/SchemaExtension/3/1#>	Please pay attention that for difference models xmlns:dm=http://iec.ch/TC57/61970-552/DifferenceModel/1# is included as well
<dm:DifferenceModel rdf:about="urn:uuid:80323db8-ea73-11e3-9df6-82687f4fc15c">	This is the ID of the model included in the instance file. Please refer to Chapter 4.3 in the CGMES. The ID to which other models (full or difference) refer in case of depend on or supersede. Please note that the “urn:uuid:” is used only for rdf:about or rdf:resource in the file header, i.e. it is not applied for other elements in the instance file.
<md:Model.scenarioTime> 2030-01-15T17:00:00.000</md:Model.scenarioTime>	This is the date & time that this model represents, i.e. for which the model is valid. The format is an extended format

	according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.created>2014-05-15T17:48:31.474</md:Model.created>	The date & time when the model was created. It is the time of the serialization. The format is an extended format according to the ISO 8601-2005. The ENTSO-E exchanges should refer to UTC.
<md:Model.description>CGMES Conformity Assessment: This is guidelines on the file header.</md:Model.description>	A description of the model and explanation of the purpose.
<md:Model.version>3</md:Model.version>	Version of the model. If the instance file is imported and exported with no change the version number is the kept same. The version changes only if the content of the file changes. It is the same logic as for the header id (see CGMES [R.4.3.4.1.]). The version is the human readable id. The version is integer number.
<md:Model.profile>http://entsoe.eu/CIM/Topology/4/1</md:Model.profile>	One to many URN/URI describing the profiles that governs this model. It uniquely identifies the profiles and its version. In this case it is the profile URI of the TP profile.
<md:Model.DependentOn rdf:resource="urn:uuid:21a419b4-ea74-11e3-9df6-82687f4fc15c" />	A reference to the model documents that the model described by this document depends on, e.g. in this case the T3 model depends on difference model E2 which has <dm:DifferenceModel rdf:about="urn:uuid:21a419b4-ea74-11e3-9df6-82687f4fc15c">. In general there can be 0 or many Model.DependentOn depending on the profile and the content of the instance file.
<md:Model.Supersedes rdf:resource="urn:uuid:7ca72efa-e952-11e3-89cf-82687f4fc15c"/>	When a model is updated the resulting model supersedes the models that were used as basis for the update. Hence this is a reference to the CIMXML documents which are superseded by this model. A model (or instance file) can supersede 1 or more models, e.g. a difference model or a full model supersede multiple models (difference or full). In this case more than one Model.Supersedes are included in the header. The referenced document(s) is (are) identified by the URN/MRID/UUID in the FullModel rdf:about attribute when full model(s) is (are) referenced and by the URN/MRID/UUID in the DifferenceModel rdf:about attribute when difference model(s) is (are) referenced. In this case the model T3 supersedes the difference model T2, i.e. this difference model is applied on the model which has the id mentioned in this Model.Supersedes.
<md:Model.modelingAuthoritySet>http://elia.be/Planning/CGMES/2.4.14</md:Model.modelingAuthoritySet>	A URN/URI referring to the organisation role / model authority set reference. The organization role is the source of the model. It is the same for all profiles part of a model exchange. This concept might be modified in the future when applying the concept of “frames” which is under discussion.
Here the place for the content of the difference file	This follows the syntax as defined in the 61970-452.
</dm:DifferenceModel>	

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Annex D
(normative)

PST transformer modelling

D.1 Introduction

This annex describes Phase-Shifting Transformer (PST) modelling according to the type and technology of the equipment; and more specifically, equivalent series reactance of phase shifters in the positive sequence as a function of the phase shift angle. In these specific models, the resistances and the magnetizing currents are always neglected.
This document also provides the mapping between the equations for each type and technology of PST to the CIM classes and attributes which are included in the CGMES.

D.2 Mapping to CIM classes and attributes

To illustrate how to use the PST modelling described in this document, the mapping to CGMES classes is presented in the following table:

PST type	**CIM class**
General Case	**cim:PhaseTapChangerTabular**
Symmetrical phase shifters	cim:PhaseTapChangerSymmetrical or cim:PhaseTapChangerLinear
Asymmetrical phase shifter	cim:PhaseTapChangerAsymmetrical
In-phase transformer and symmetrical phase shifter	cim:PhaseTapChangerSymmetrical and cim:RatioTapChanger
In-phase transformer and asymmetrical phase shifter	cim:PhaseTapChangerAsymmetrical and cim:RatioTapChanger

Recommendation: It is highly recommended to use tabular data to exchange PST parameters (cim:PhaseTapChangerTabular) instead of having to recalculate the parameters per tap according to each model type

PST attribute	**CIM attribute**
n	cim:PhaseTapChangerTablePoint.step
n_0	cim:TapChanger.neutralStep
δu	cim:PhaseTapChangerNonLinear.voltageStepIncrement
r	cim:TapChangerTablePoint.ratio
α	cim:PhaseTapChangerTablePoint.angle
$\delta\alpha$	cim:PhaseTapChangerLinear.stepPhaseShiftIncrement
$X(\alpha)$	cim:TapChangerTablePoint.x
$X(0)$	cim:PhaseTapChangerLinear.xMin or cim:PhaseTapChangerNonLinear.xMin
$X(\alpha_{\max})$	cim:PhaseTapChangerLinear.xMax or cim:PhaseTapChangerNonLinear.xMax

θ	cim:PhaseTapChangerAsymmetrical.windingConnectionAngle

Notes:

- Angle sign convention: Positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer).
- RatioTapChanger.stepVoltageIncrement: Both positive and negative values are allowed.
- PhaseTapChangerNonLinear.voltageStepIncrement: Both positive and negative values are allowed.
- PhaseTapChangerAsymmetrical.windingConnectionAngle: Both positive and negative values are allowed.

D.3 Reactance formulas summary table

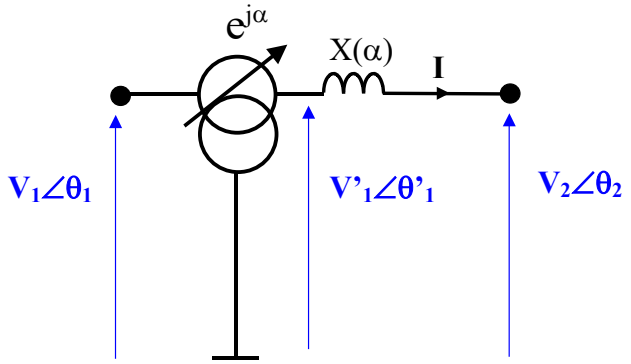
Equipment type	Equivalent series reactance as a function of the phase shift angle
Symmetrical Phase shifters	$X(\alpha) = X(0) + \left(X(\alpha_{\max}) - X(0)\right)\left(\frac{\sin(\alpha/2)}{\sin(\alpha_{\max}/2)}\right)^2$
Asymmetrical Phase Shifter	$X(\alpha) = X(0) + \left(X(\alpha_{\max}) - X(0)\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\frac{\sin\theta - \tan\alpha_{\max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$
In-phase transformer and symmetrical phase shifter	$X(r,\alpha) = X_r(r_{nom})\left(\frac{r}{r_{nom}}\right)^2 + X_\alpha(0) + \left(X_\alpha(\alpha_{\max}) - X_\alpha(0)\right)\left(\frac{\sin(\alpha/2)}{\sin(\alpha_{\max}/2)}\right)^2$
In-phase transformer and asymmetrical phase shifter	$X(r,\alpha) = X_r(r_{nom})\left(\frac{r}{r_{nom}}\right)^2 + X_\alpha(0)$ $+ \left(X_\alpha(\alpha^0_{\max}) - X_\alpha(0)\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}(r)}\frac{\sin\theta - \tan\alpha_{\max}(r)\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$ with $\alpha_{\max}(r) = A\tan\left(\frac{\sin\theta}{\frac{r}{r_{nom}\tan\alpha^0_{\max}}\left(\sin\theta - \tan\alpha^0_{\max}\cos\theta\right) + \cos\theta}\right)$ and $\alpha^0_{\max} = \alpha_{\max}(r_{nom})$

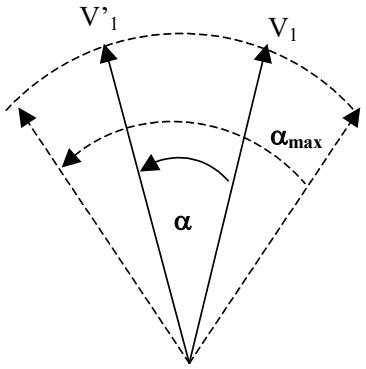
Variable	Meaning
$\alpha_{\max}$	Maximal phase shift: the maximum angle value of the phase shifter angles table
r_{nom}	Nominal ratio of the in-phase transformer
$\alpha^0_{\max}$	Maximal phase shift when the in-phase transformer ratio is nominal

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D.4 Symmetrical Phase shifters

D.4.1 One phase diagram and equations





$$V2 = V1e^{j\alpha} - jX(\alpha)I \qquad V'1 = V1\left(1 + 2j\sin\frac{\alpha}{2}e^{j\frac{\alpha}{2}}\right) = e^{j\alpha}V1$$

D.4.2 Expression of the angle and ratio per tap

Based on the figure above:

$$\alpha = (n - n_0)\cdot\delta\alpha \qquad \text{or} \qquad \alpha = 2A\tan\left(\frac{(n - n_0)\cdot\delta u}{2}\right)$$

$$r = 1$$

D.4.3 Expression of the equivalent series reactance given the angle

Assuming the reactance of the regulating winding varies as the square of the number of turns, the equivalent reactance can be written as follows for non delta-hexagonal technologies (see proof in section D.9.1):

$$X(\alpha) = X(0) + \left(X(\alpha_{\max}) - X(0)\right)\left(\frac{\sin(\alpha/2)}{\sin(\alpha_{\max}/2)}\right)^2$$

3 parameters:

- α_{max}: maximal phase shift
- X(0): equivalent series reactance at zero phase shift $\alpha=0$
- $X(\alpha_{max})$: equivalent series reactance at maximal phase shift $\alpha=\alpha_{max}$

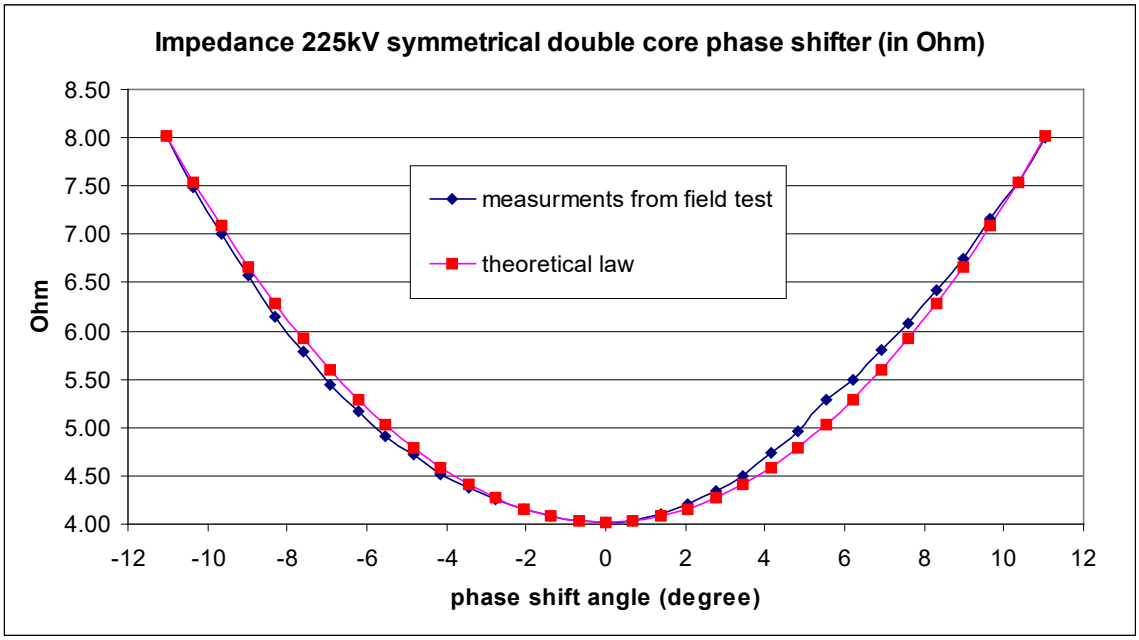
1 variable:

- α: current phase shift

The formula above is valid for single or double core symmetrical phase shifters except for the hexagonal technology.

633 For single core symmetrical phase shifters: X(0)=0

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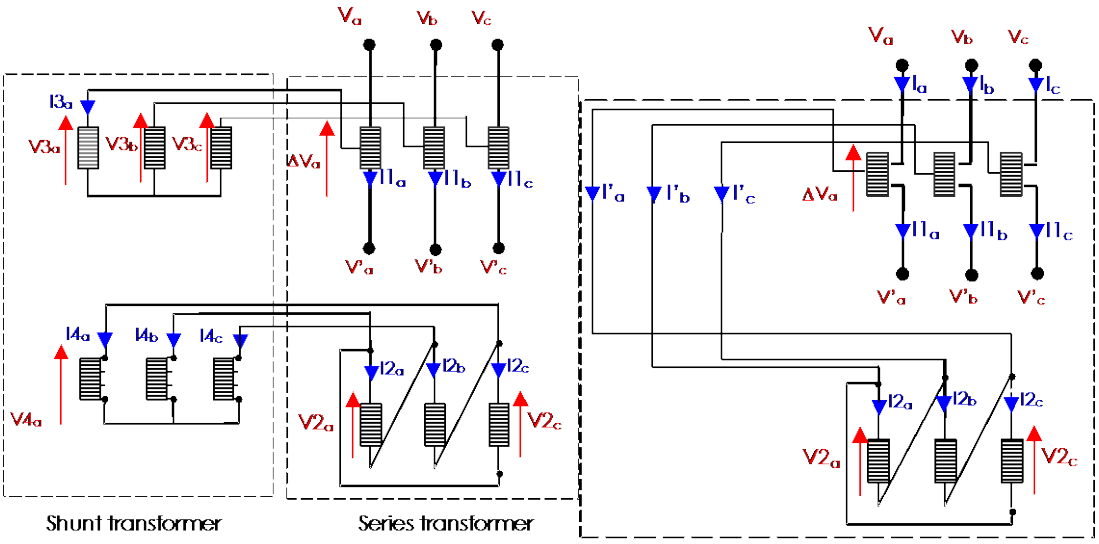


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637 ### D.4.4 Three-phase diagrams

638 Dual core: Single core:



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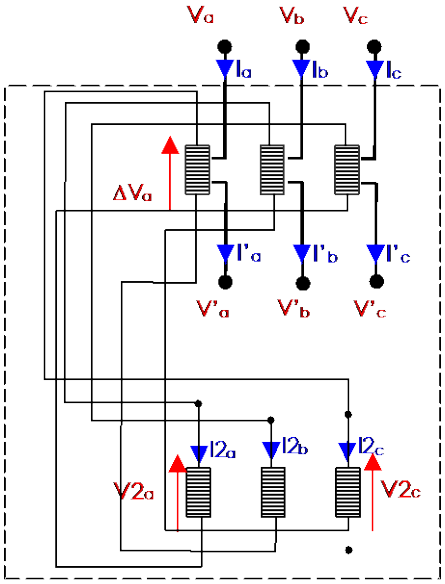
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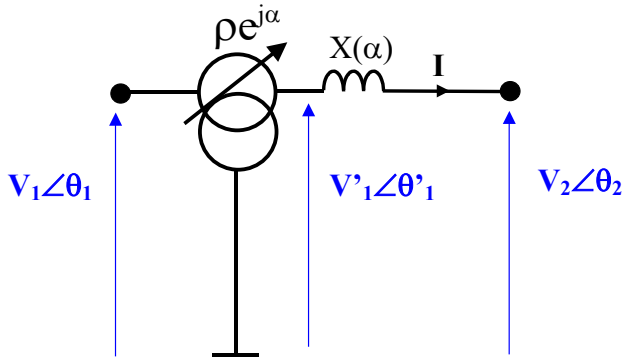
643

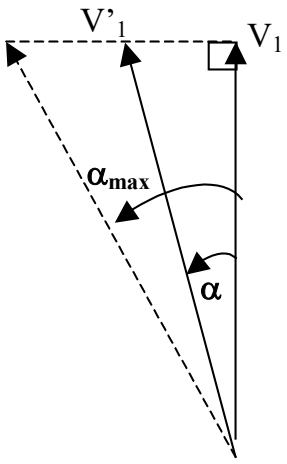
644 Single core, delta hexagonal:



D.5 Quadrature booster

D.5.1 One phase diagram and equations





$$V'1 = V1\left(1 + j\tan\alpha\right) = \rho e^{j\alpha} V1 \qquad V2 = V1\rho e^{j\alpha} - jX(\alpha)I$$

ρ and α vary with ρ.cos α = constant.

D.5.2 Expression of the angle and ratio per tap

Based on the figure above:

$$\alpha = A\tan\left((n - n_0)\cdot\delta u\right)$$

$$r = \frac{1}{\sqrt{\left((n - n_0)\cdot\delta u\right)^2 + 1}}$$

D.5.3 Expression of the equivalent series reactance given the angle

Assuming the reactance of the regulating winding varies as the square of the number of turns, the equivalent reactance of the quadrature booster can be written as follows (see proof in section D.9.2):

$$X(\alpha) = X(0) + \left(X(\alpha_{max}) - X(0)\right)\left(\frac{\tan(\alpha)}{\tan(\alpha_{max})}\right)^2$$

3 parameters:

- α_{max}: maximal phase shift
- X(0): equivalent series reactance at zero phase shift
- $X(\alpha_{max})$: equivalent series reactance at maximal phase shift

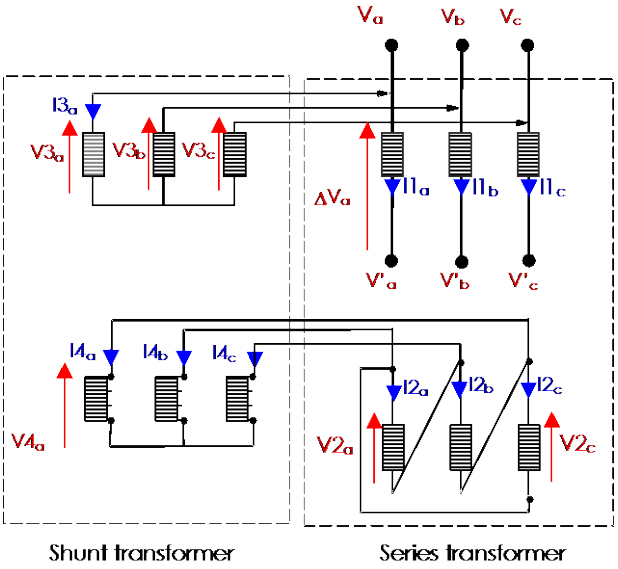
1 variable:

- α: current phase shift

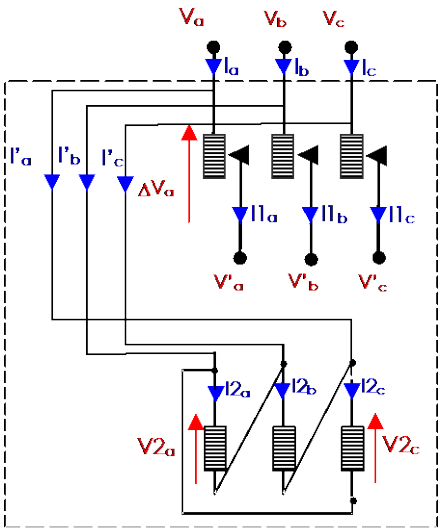
For quadrature boosters with a single core: X(0)=0

D.5.4 Three-phase diagrams

Dual core:

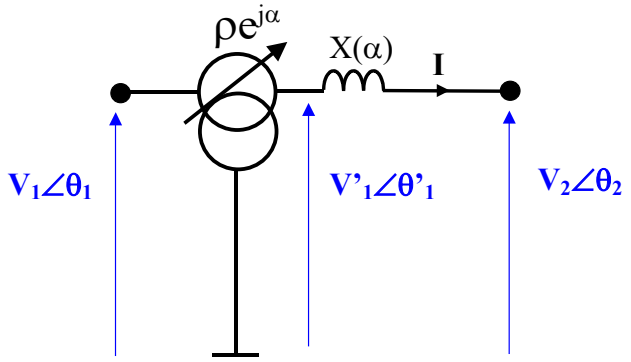


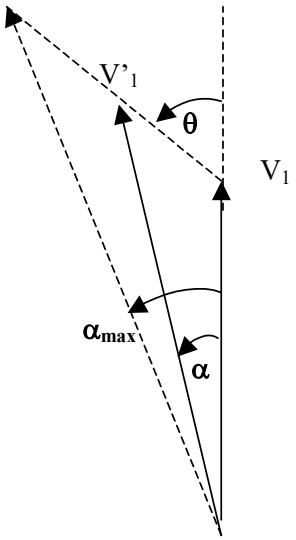
Single core:



D.6 Asymmetrical Phase Shifter

D.6.1 One phase diagram and equations





$$V'1 = V1\left(1 + e^{j\theta}.\frac{\tan\alpha}{\sin\theta - \tan\alpha\cos\theta}\right) = \rho e^{j\alpha}V1 \qquad V2 = V1.\rho e^{j\alpha} - jX(\alpha)I$$

θ is fixed, only α and ρ vary.

D.6.2 Expression of the angle and ratio per tap

Based on the figure above:

$$\alpha = A\tan\left(\frac{(n-n_0)\cdot\delta u\cdot\sin\theta}{1+(n-n_0)\cdot\delta u\cdot\cos\theta}\right)$$

$$r = \frac{1}{\sqrt{\left((n-n_0)\cdot\delta u\cdot\sin\theta\right)^2 + \left(1+(n-n_0)\cdot\delta u\cdot\sin\theta\right)^2}}$$

D.6.3 Expression of the equivalent series reactance given the angle

Assuming the reactance of the regulating winding varies as the square of the number of turns, the equivalent reactance can be written as follows (see proof in section D.9.3):

$$X(\alpha) = X(0) + \left(X(\alpha_{\max}) - X(0)\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\frac{\sin\theta - \tan\alpha_{\max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$$

4 parameters:

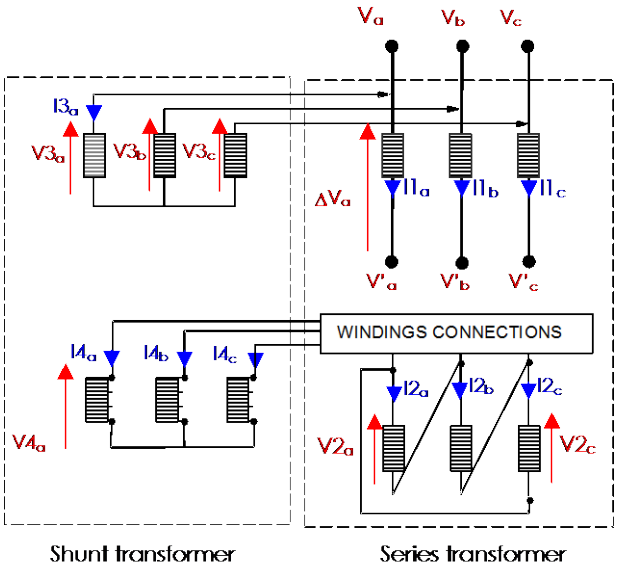
- $\alpha_{max:}$ maximal phase shift
- X(0): equivalent series reactance at zero phase shift
- $X(\alpha_{max})$: equivalent series reactance at maximal phase shift
- θ: boost voltage angle

1 variable:

- $\alpha_{:}$ current phase shift

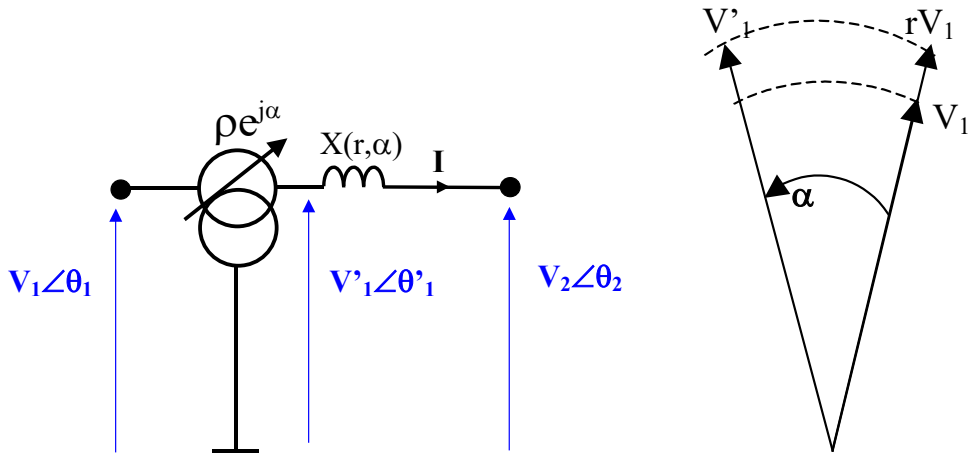
D.6.4 Three-phase diagram

Dual core:



D.7 In-phase transformer and symmetrical phase shifter

D.7.1 One phase diagram and equations



$$V2 = \rho V1 e^{j\alpha} - jX(\alpha)I. \qquad V'1 = r.V1\left(1 + 2j\sin\frac{\alpha}{2}e^{j\frac{\alpha}{2}}\right) = \rho e^{j\alpha}V1$$

r and α vary, and ρ=r

D.7.2 Expression of the angle and ratio per tap

Same as D.4.2 with the addition of the in-phase transformer ratio r.

D.7.3 Expression of the equivalent series reactance given the angle and the in-phase transformer ratio

Assuming:

- the reactance of the regulating winding varies as the square of the number of turns,
- the equivalent reactance is the sum of the reactance of the in-phase transformer X_r and the reactance of the phase shifter part X_α,
- the phase shifting angle α does not depend on the in-phase ratio r

The equivalent reactance can be written as follows (**not proved**):

$$X(r,\alpha) = X_r(r_{nom})\left(\frac{r}{r_{nom}}\right)^2 + X_\alpha(0) + \left(X_\alpha(\alpha_{\max}) - X_\alpha(0)\right)\left(\frac{\sin(\alpha/2)}{\sin(\alpha_{\max}/2)}\right)^2$$

6 Parameters:

- r_{nom}: nominal ratio of the in-phase transformer
- $X_r(r_{nom})$: equivalent series reactance of the in-phase transformer at nominal in-phase ratio
- α_{max}: maximal phase shift
- $X_\alpha(0)$: equivalent series reactance of the phase shifter part at zero phase shift
- $X_\alpha(\alpha_{max})$: equivalent series reactance of the phase shifter part at maximal phase shift at nominal in-phase ratio (r_{nom})

2 Variables:

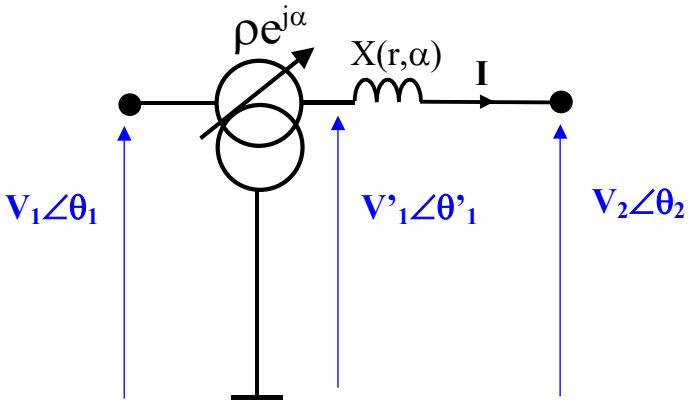
- r: current ratio of the in-phase transformer
- α: current phase shift

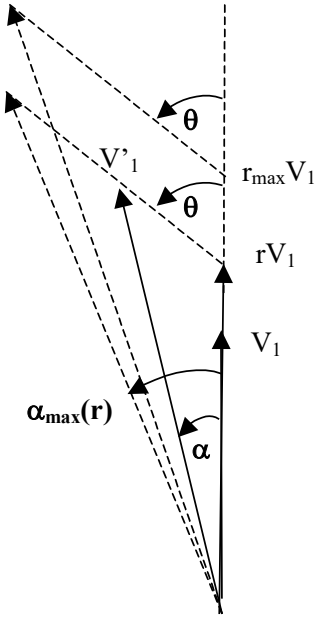
D.7.4 Technology principles

No example found.

D.8 In-phase transformer and asymmetrical phase shifter

D.8.1 One phase diagram and equations





$$V'1 = rV1\left(1 + e^{j\theta}.\frac{\tan\alpha}{\sin\theta - \tan\alpha\cos\theta}\right) = \rho e^{j\alpha}V1 \qquad V2 = \left(V1 - jX(\alpha)I\right).\rho e^{j\alpha}$$

θ is fixed, only r and α are variables, ρ varies as a consequence.

D.8.2 Expression of the angle and ratio per tap

Not proved.

D.8.3 Expression of the equivalent series reactance given the angle and the in-phase transformer ratio

Assuming:

- the reactance of the regulating winding varies as the square of the number of turns,
- the equivalent reactance is the sum of the reactance of the in-phase transformer X_r and the reactance of the phase shifter part X_α,

The equivalent series reactance can be written as follows (**not proved**):

$$X(r,\alpha) = X_r(r_{nom})\left(\frac{r}{r_{nom}}\right)^2 + X_\alpha(0) + \left(X_\alpha(\alpha^0_{\max}) - X_\alpha(0)\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}(r)}\,\frac{\sin\theta - \tan\alpha_{\max}(r)\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$$

with $\alpha_{\max}(r) = A\tan\left(\frac{\sin\theta}{\frac{r}{r_{nom}\tan\alpha^{0}_{\max}}\left(\sin\theta - \tan\alpha^{0}_{\max}\cos\theta\right)+\cos\theta}\right)$ with $\alpha^{0}_{\max} = \alpha_{\max}(r_{nom})$

6 Parameters:

- r_{nom}: nominal ratio of the in-phase transformer
- $X_r(r_{nom})$: equivalent series reactance of the in-phase transformer at nominal in-phase ratio r_{nom}
- θ: fix boost voltage angle
- $\alpha^{0}_{\max} = \alpha_{\max}(r_{nom})$: maximal phase shift for nominal in-phase ratio (r_{nom})
- $X_\alpha(0)$: equivalent series reactance of the phase shifter part at zero phase shift
- $X_\alpha(\alpha^0_{max})$: equivalent series reactance of the phase shifter part at maximal phase shift at nominal in-phase ratio (r_{nom})

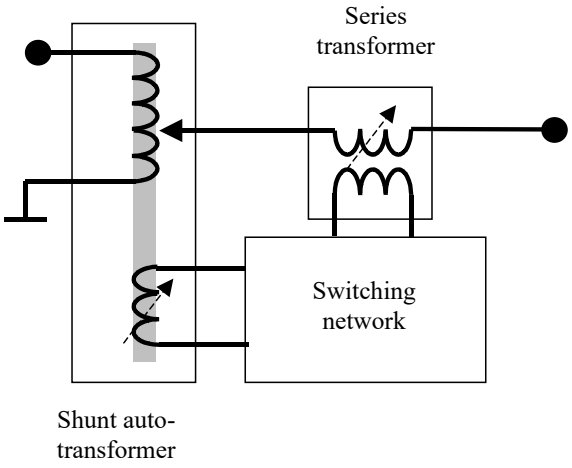
2 Variables:

- r: current ratio of the in-phase transformer
- α: current phase shift

For $\theta = \pi/2$ (quadrature booster): $\alpha_{\max}(r) = A\tan\left(\frac{r_{nom}}{r}\tan\alpha^{0}_{\max}\right)$

D.8.4 Technology principles

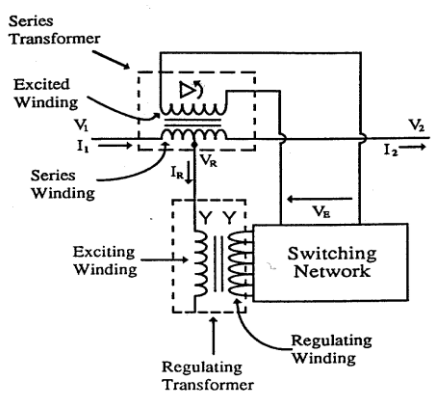
In-phase regulating auto-transformer

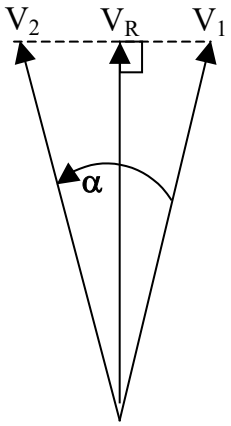


The phase shift regulation may be on the shunt or on the series transformer

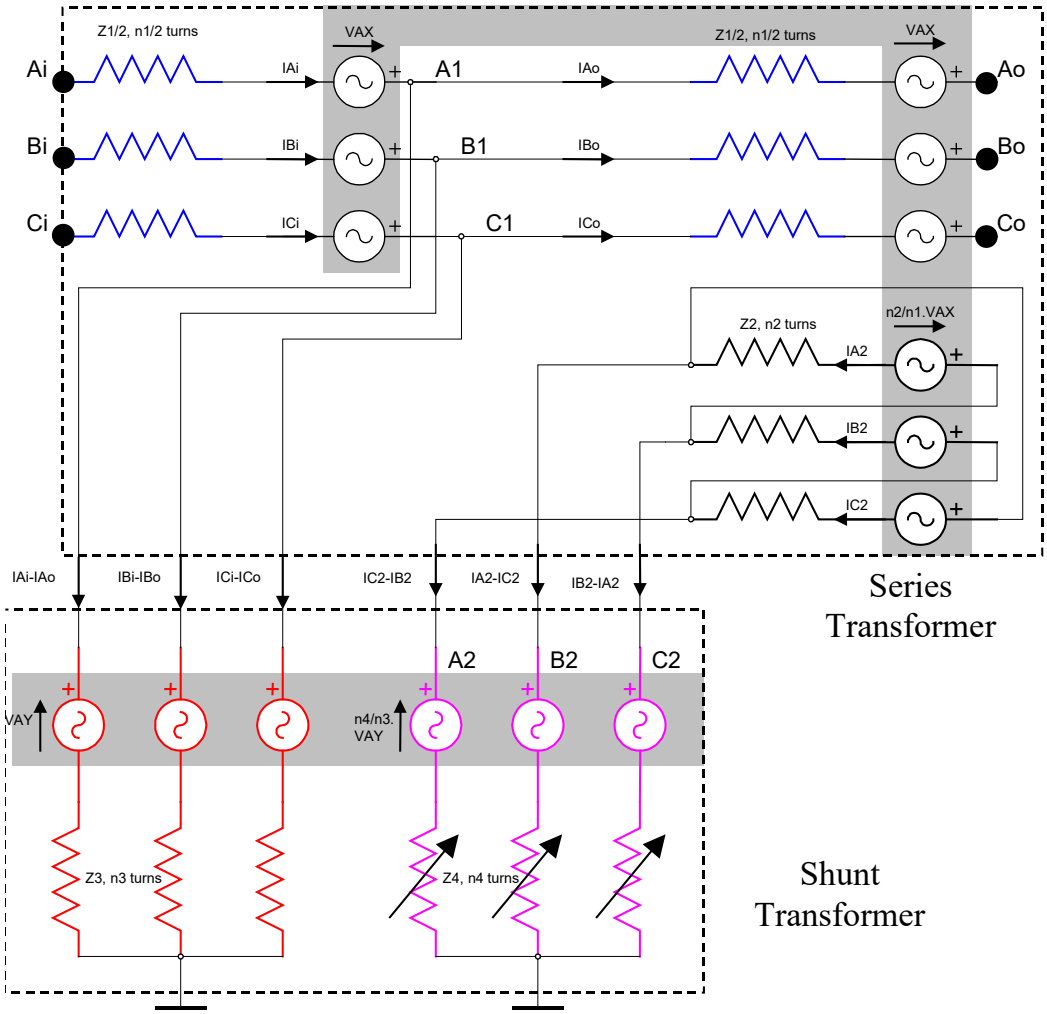
D.9 Detailed calculations and examples

D.9.1 Symmetrical Phase Shifters with two cores





Detailed three phase diagram



788

789 Only n4 varies.

790 Example of numerical values:

791 n1=120

792 n2=318

793 n3=680

794 $n4_{max}$=182

795 X1=1,23 Ω

796 X2=8,64 Ω

797 X3=101 Ω

798

799 **Expression of the output current Io and the shunt current I3=Ii-Io:**

800 When considering an ideal phase shifter, the conservation of the electric power is written:
801 $Si = 3ViIi^* = 3VoIo^*$

802 As the voltage angle is shifted by alpha: $Vo = e^{j\alpha}Vi$

803 The current angle must also be shifted by alpha as well: $\boxed{Io = e^{j\alpha}Ii}$

804 Then, the shunt current $\boxed{I3 = Ii - Io = (1 - e^{j\alpha})Ii}$

805

806 **Expression of the shunt equivalent reactance**

807

808 The shunt reactance Xshunt(α) is defined as the equivalent reactance which crossed by the series input
809 current (*Ii*) would produce the reactive losses of the shunt transformer:

810 $$Qshunt = 3.Xshunt(\alpha).|Ii|^2$$

811 with:

812 $$Qshunt = 3.X3.|I3|^2 + 3.X4.|I4|^2$$

813 $$|I3|^2 = |Ii|^2|1 - e^{j\alpha}|^2 = 2(1 - \cos\alpha)|Ii|^2 = 4(\sin(\alpha/2))^2$$

814 $$|I4|^2 = \frac{n3^2}{n4^2}|I3|^2$$

$$Xshunt(\alpha) = 4\left(\sin(\alpha/2)\right)^2\left(X3 + X4\left(\frac{n3}{n4}\right)^2\right)$$

Hypothesis: $X4 = X4_{\max}\left(\frac{n4}{n4_{\max}}\right)^2$

Then $Xshunt(\alpha) = 4\left(\sin\left(\alpha/2\right)\right)^2\left(X3 + X4_{\max}\frac{n3^2}{n4_{\max}^2}\right)$

Expression of the equivalent series reactance

The series reactance Xseries(α) is defined as the equivalent reactance which crossed by the series input current (*Ii*) would produce the reactive losses of the series transformer:

$$Qseries = 3.Xseries(\alpha).|Ii|^2$$

with:

$$Qseries = 3.\frac{X1}{2}.|Ii|^2 + 3.\frac{X1}{2}.|Io|^2 + 3.X2.|I2|^2$$

as seen previously: $Io = e^{j\alpha}Ii$

series transformer current relationship: $\frac{n1}{2}.Ii + \frac{n1}{2}.Io = +n2.I2$

then $I2 = \frac{n1}{2n2}\left(1 + e^{j\alpha}\right)Ii$ and $|I2|^2 = \left(\frac{n1}{2n2}\right)^2|Ii|^2\left|1 + e^{j\alpha}\right|^2 = \frac{1}{2}\frac{n1^2}{n2^2}(1+\cos\alpha)|Ii|^2$

then

$$Xseries(\alpha) = X1 + \frac{1}{2}\left(\frac{n1}{n2}\right)^2(1+\cos\alpha)X2 = X1 + \left(\frac{n1}{n2}\right)^2(1-\sin(\alpha/2))X2$$

$$Xseries(\alpha) = \left(X1 + \left(\frac{n1}{n2}\right)^2 X2\right) - \left(\frac{n1}{n2}\right)^2 X2\left(\sin(\alpha/2)\right)^2$$

Expression of the total equivalent reactance X

835 $$X(\alpha)=\left(X1+\left(\frac{n1}{n2}\right)^2 X2\right)+4\left(\sin(\alpha/2)\right)^2\left(X3+X4_{\max}\frac{n3^2}{n4_{\max}^2}-\left(\frac{n1}{2n2}\right)^2 X2\right)$$

836

837 or

838 $$X(\alpha)=X(0)+\left(X(\alpha_{\max})-X(0)\right)\left(\frac{\sin(\alpha/2)}{\sin(\alpha_{\max}/2)}\right)^2$$

839 with:

840 $$X(0)=\left(X1+\left(\frac{n1}{n2}\right)^2 X2\right)$$

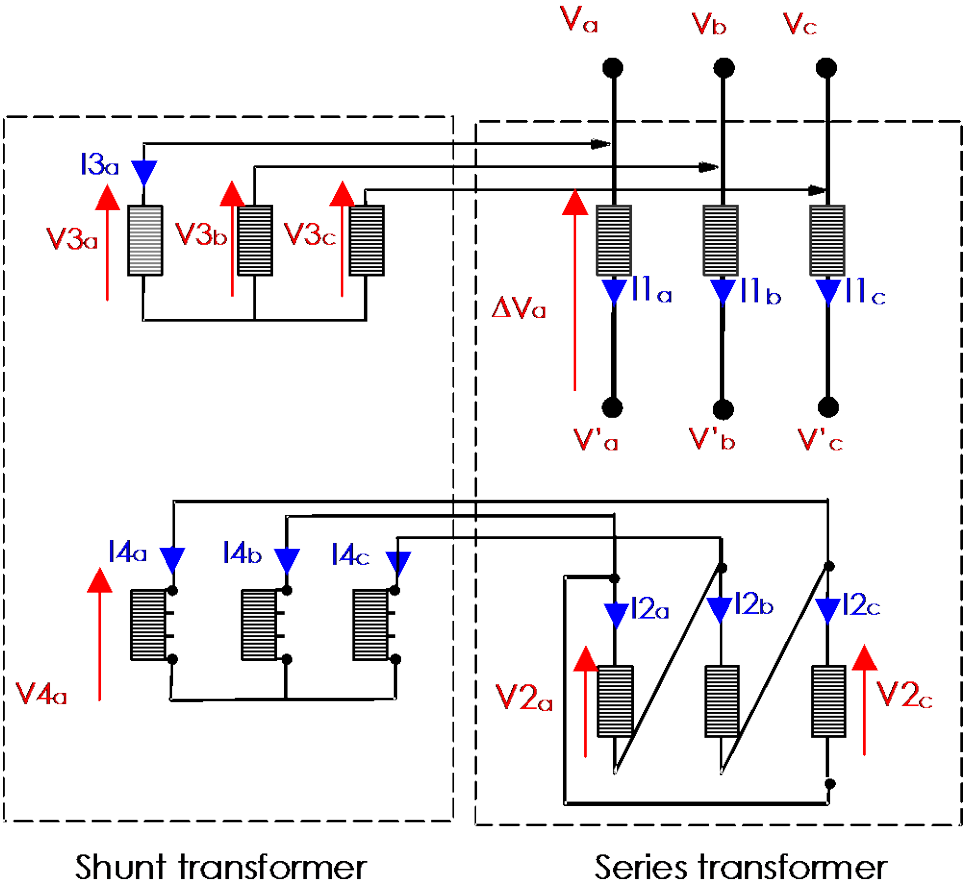
841 $$X(\alpha_{\max})-X(0)=4\left(\sin(\alpha_{\max}/2)\right)^2\left(X3+X4_{\max}\frac{n3^2}{n4_{\max}^2}-\left(\frac{n1}{2n2}\right)^2 X2\right)$$

842 **Remark:** only α varies.

843 ## D.9.2 Quadrature boosters

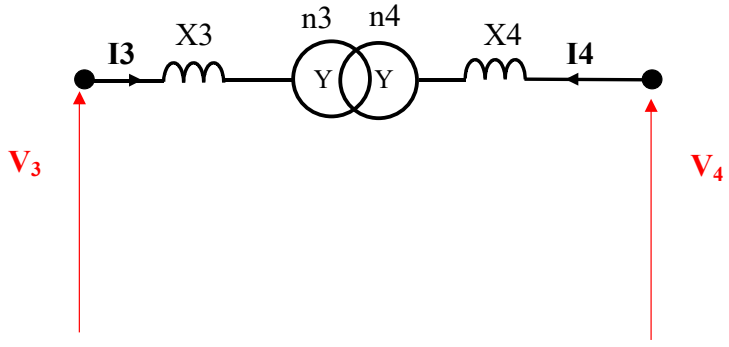
844 ### D.9.2.1 Quadrature booster with two cores

845 **DETAILED THREE PHASE DIAGRAM**



For each physical value X (X=V or I) $X = X_a = aX_b = a^2X_c$ with $a = e^{j\frac{2\pi}{3}}$

shunt transformer with variable ratio n4/n3 with X3 reactance on the primary winding and X4 on the secondary winding:



$$V4 - jX4.I4 = \frac{n4}{n3}(V3 - jX3.I3) \qquad I3 = -\frac{n4}{n3}I4$$

$$V4 = \frac{n4}{n3}V3 + j\left(X_4 + X_3\left(\frac{n4}{n3}\right)^2 \right) I4$$

Hypothesis: the X4 reactance varies like the square of the number of turns n4:

$$X_4 = X_4^{\max}\left(\frac{n4}{n4_{\max}}\right)^2$$

Series transformer of fixed ratio n1/n2 with X1 reactance on the primary winding and X2 on the secondary winding:

$$\Delta V = \frac{n1}{n2}V2 + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2 \right) I1$$

$$I2 = -\frac{n1}{n2}I1$$

Link between series and shunt transformers:

- Primary windings:
  $V3 = V$
- Secondary windings:
  $$I4 = I4_a = I2_b - I2_c = (a^2 - a)I2_a = -j\sqrt{3}I2$$
  $$V2 = V2_a = V4_b - V4_c = (a^2 - a)V4 = -j\sqrt{3}V4$$

Calculation:

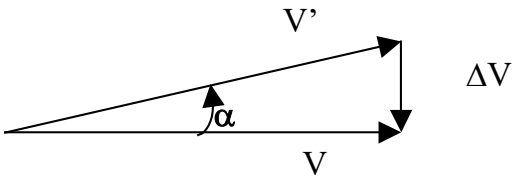
$$\Delta V = -j\frac{n4.n1}{n3.n2}\sqrt{3}V + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2 + 3\left(\frac{n1}{n2}\right)^2 n4^2\left(\frac{X4^{\max}}{n4_{\max}^2} + \frac{X3}{n3^2}\right)\right) I1$$

$$I3 = -j\frac{n4.n1}{n3.n2}\sqrt{3}I1$$

At no load conditions (I=0):

$$\Delta V = -j\frac{n4.n1}{n3.n2}\sqrt{3}V$$

And geometrically:



$$\Delta V = -jV\tan\alpha$$

hence $\frac{n4}{n4^{\max}} = \frac{\tan\alpha}{\tan\alpha_{\max}}$

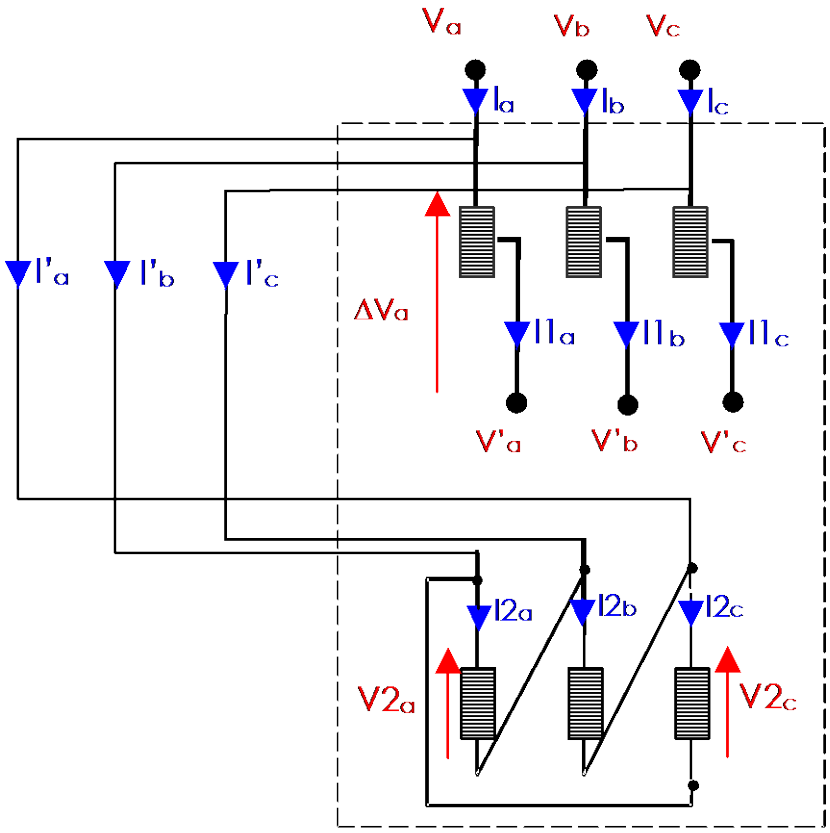
And $\Delta V = -j\frac{n4.n1}{n3.n2}\sqrt{3}V + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2 + 3n4_{\max}^2\left(\frac{n1}{n2}\right)^2\left(\frac{X4^{\max}}{n4_{\max}^2} + \frac{X3}{n3^2}\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\right)^2\right)I1$

comes:

$$\Delta V = -j\frac{n4.n1}{n3.n2}\sqrt{3}V + jXI \quad \text{with} \quad \boxed{X(\alpha) = X(0) + \left(X(\alpha_{\max}) - X(0)\right)\left(\frac{\tan(\alpha)}{\tan(\alpha_{\max})}\right)^2}$$

D.9.2.2 Quadrature booster with a single core

Detailed three phase diagram



n1 varies.

$$I' = j\sqrt{3}I2$$

$$V2 = -j\sqrt{3}V$$

$$\Delta V = \frac{n1}{n2}V2 + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2\right)I1$$

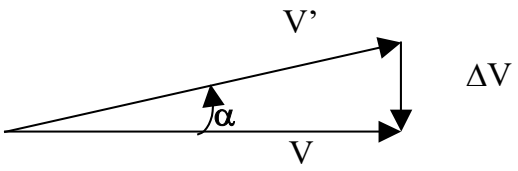
$$\Delta V = -j\sqrt{3}\frac{n1}{n2}V + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2\right)I1$$

Assuming $X1 = X1_{\max}\left(\frac{n1}{n1_{\max}}\right)^2$

$$\Delta V = -j\sqrt{3}\frac{n1}{n2}V + j\left(\frac{X1_{\max}}{n1_{\max}} + \frac{X2}{n2^2}\right)n1^2 I1$$

900 At no-load: $\Delta V = -j\sqrt{3}\frac{n1}{n2}V$

901 And geometrically:



902 $\Delta V = -jV\tan\alpha$

903

904 hence $\frac{n1}{n1^{\max}} = \frac{\tan\alpha}{\tan\alpha_{\max}}$

905 $X(\alpha) = \left(X1_{\max} + X2\frac{n1_{\max}^2}{n2^2}\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\right)^2$

906 $X(\alpha) = X(\alpha_{\max})\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\right)^2$

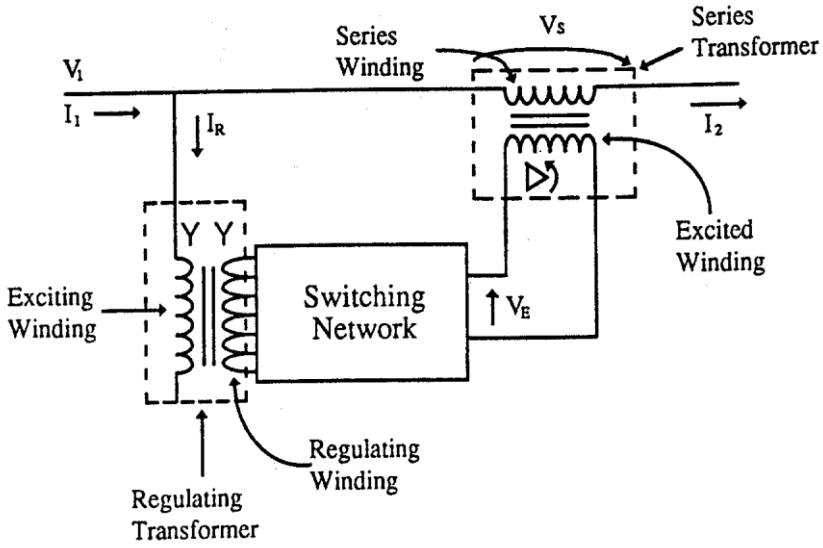
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910 **D.9.3 Asymmetrical phase shifter**

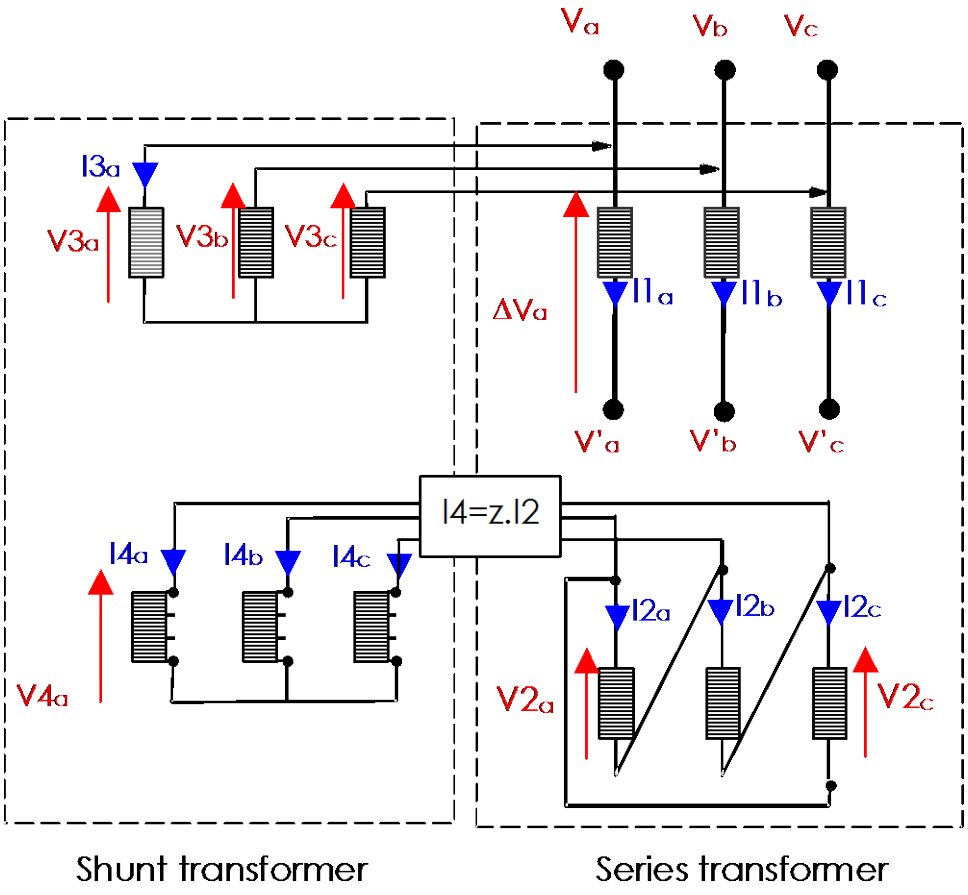
911 **D.9.3.1 Asymmetrical phase shifter with two cores**



912

913

914 Detailed three phase diagram:



915

For each physical value X (X=V or I) $X = X_a = aX_b = a^2 X_c$ with $a = e^{j\frac{2\pi}{3}}$

shunt transformer equations:

$$V4 = \frac{n4}{n3} V3 + j\left(X_4 + X_3 \left(\frac{n4}{n3} \right)^2 \right) I4$$

$$I3 = -\frac{n4}{n3} I4$$

Hypothesis: the X_2 reactance varies like the square of the k_1 ratio:

$$X_4 = X_4^{\max} \left(\frac{n4}{n4_{\max}} \right)^2$$

Series transformer of fixed ratio k_2 with X_{series} reactance on the primary winding:

$$\Delta V = \frac{n1}{n2} V2 + j\left(X1 + \left(\frac{n1}{n2} \right)^2 X2 \right) I1$$

$$I2 = -\frac{n1}{n2} I1$$

Link between series and shunt transformers:

- Primary windings:
  $V3 = V$
- "z", a complex number, determines the coupling between the series and shunt transformer secondary windings:
  $I4 = I4_a = z.I2$

  $V2 = z.V4$

  for a quadrature booster $z = -j\sqrt{3}$

Calculation:

939 $$\Delta V = \frac{n4.n1}{n3.n2}.z.V + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2 - z^2\left(\frac{n1}{n2}\right)^2 n4^2\left(\frac{X4^{\max}}{n4_{\max}^2} + \frac{X3}{n3^2}\right)\right) I1$$

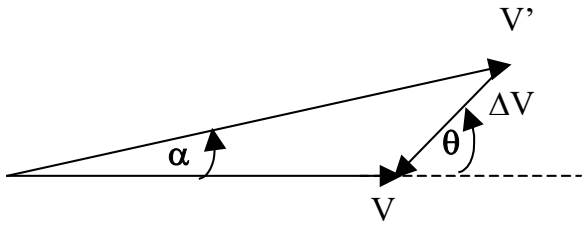
940 $$I3 = \frac{n4.n1}{n3.n2}.z.I1$$

941

942 At no load conditions (I=0):

943 $$\Delta V = \frac{n4.n1}{n3.n2}.z.V$$

944 And geometrically: $\Delta V = -e^{j\theta}.V\,\frac{\tan\alpha}{\sin\theta - \tan\alpha\cos\theta}$



945

946

947 hence $$\frac{n4}{n4^{\max}} = \frac{\tan\alpha}{\tan\alpha_{\max}}\,\frac{\sin\theta - \tan\alpha_{\max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}$$

948 And

949 $$\Delta V = -j\frac{n4.n1}{n3.n2}\sqrt{3}V + j\left(X1 + \left(\frac{n1}{n2}\right)^2 X2 - z^2 n4_{\max}^2\left(\frac{n1}{n2}\right)^2\left(\frac{X4^{\max}}{n4_{\max}^2} + \frac{X3}{n3^2}\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\,\frac{\sin\theta - \tan\alpha_{\max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2\right) I1$$

950

951

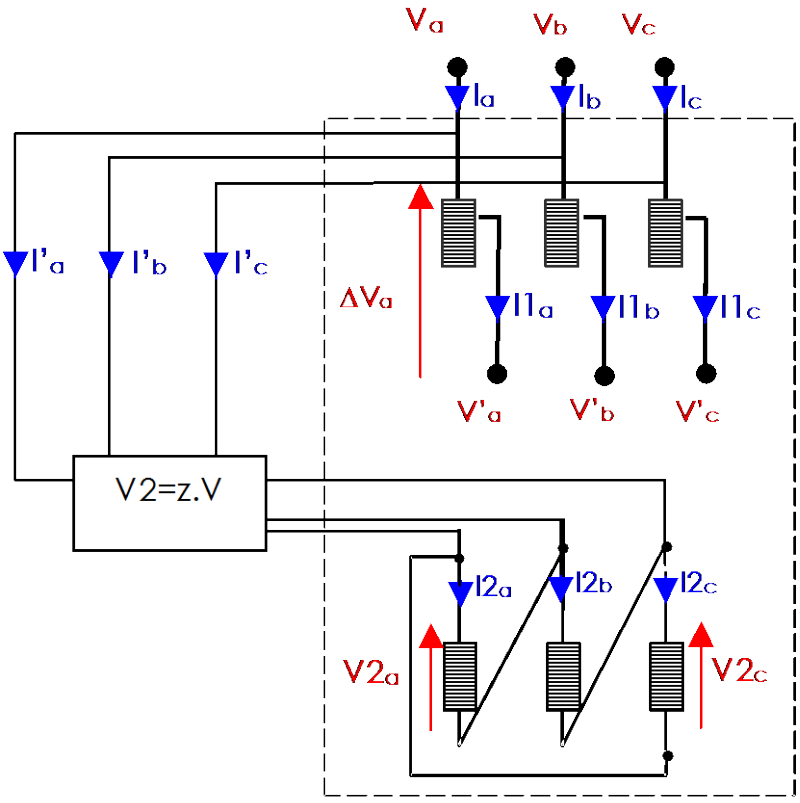
952 comes:

953

954 $$\Delta V = z\frac{n4.n1}{n3.n2}V + jXI \qquad \text{with}$$

955 $$X(\alpha) = X(0) + \left(X(\alpha_{\max}) - X(0)\right)\left(\frac{\tan\alpha}{\tan\alpha_{\max}}\,\frac{\sin\theta - \tan\alpha_{\max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$$

956 **D.9.3.2 Asymmetrical phase shifter with a single core**



957

958 Only n1 varies.

959

960 $I' = -z.I2$

961 $V2 = z.V$

962 $$\Delta V = \frac{n1}{n2} V2 + j\left(X1 + \left(\frac{n1}{n2} \right)^2 X2 \right) I1$$

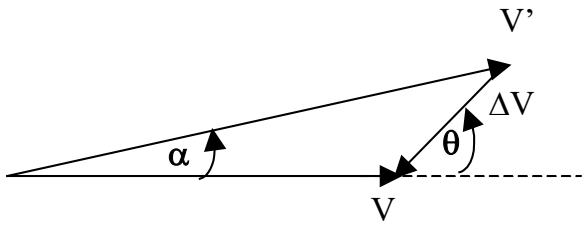
963 $$\Delta V = z.\frac{n1}{n2} V + j\left(X1 + \left(\frac{n1}{n2} \right)^2 X2 \right) I1$$

964 Assuming $X1 = X1_{\max} \left(\frac{n1}{n1_{\max}} \right)^2$

965 $$\Delta V = z.\frac{n1}{n2} V + j\left(\frac{X1_{\max}}{n1_{\max}} + \frac{X2}{n2^2} \right) n1^2 I1$$

966 At no-load: $\Delta V = z.\frac{n1}{n2} V$

967 And geometrically: $\Delta V = -e^{j\theta}.V\frac{\tan\alpha}{\sin\theta - \tan\alpha\cos\theta}$



968

969

970 hence $\frac{n1}{n1_{max}} = \frac{\tan\alpha}{\tan\alpha_{max}}\frac{\sin\theta - \tan\alpha_{max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}$

971 $X(\alpha) = \left(X1_{max} + X2\frac{n1_{max}^2}{n2^2}\right)\left(\frac{\tan\alpha}{\tan\alpha_{max}}\frac{\sin\theta - \tan\alpha_{max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$

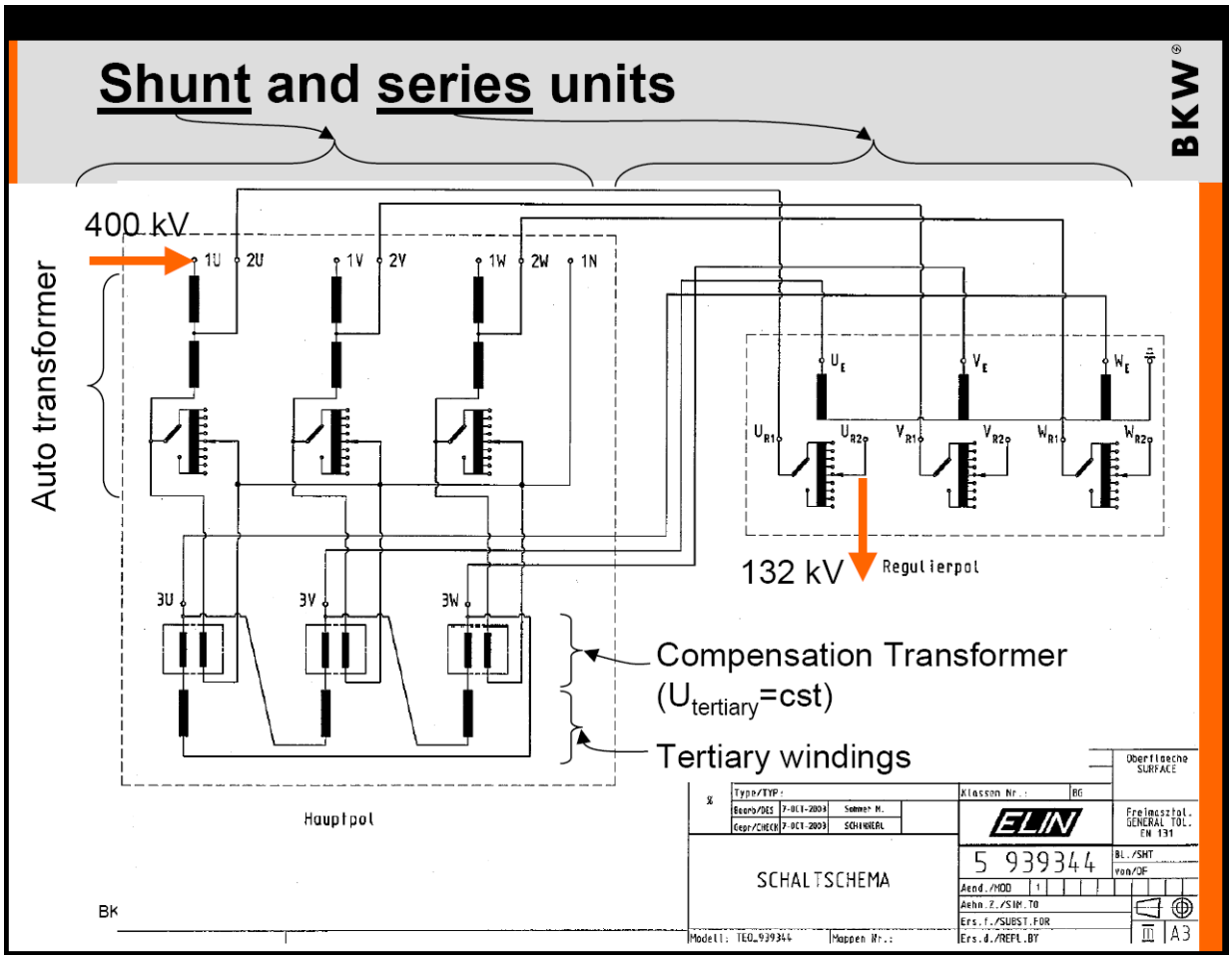
972 $X(\alpha) = X(\alpha_{max})\left(\frac{\tan\alpha}{\tan\alpha_{max}}\frac{\sin\theta - \tan\alpha_{max}\cos\theta}{\sin\theta - \tan\alpha\cos\theta}\right)^2$

973

974 **D.9.3.3 In-phase transformer and asymmetrical phase shifter**

975 Examples of detailed three-phase diagrams: voltage regulating auto-transformer and Quadrature booster:

976



977

978

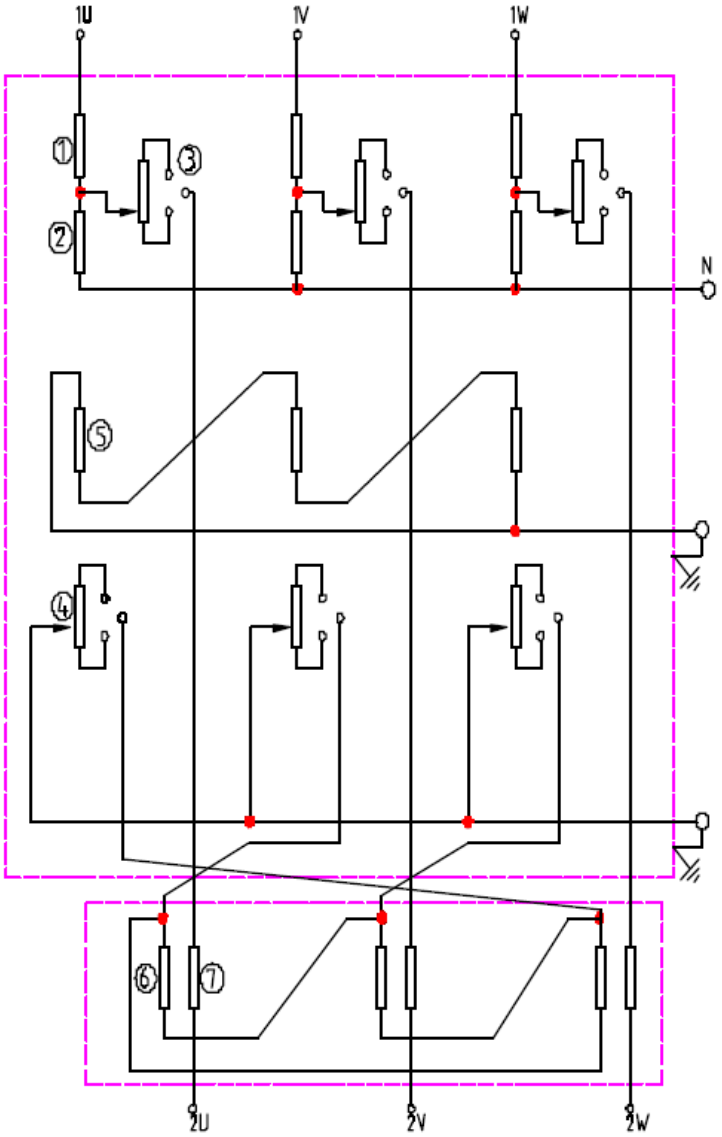
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983 Voltage regulating auto-transformer and Quadrature booster:



984

985

986 1 EHV winding
987 2 HV winding
988 3 EV voltage regulating winding
989 4 Phase shift regulating winding
990 5 Tertiary winding
991 6 Primary winding of the series transformer
992 7 Secondary winding of the series transformer
993

994

995

Annex E
(normative)

Implementation guide

E.1 Introduction

This annex provides guidelines for Suppliers and TSOs that are implementing CGMES. The need to produce such document is justified by the fact that during the implementation of the CGMES since Dec 2013 there were some issues reported but were not fully clarified in the last released version of the CGMES, i.e. version 2.4.15.

The CGMES conformity process and the tests that are performed by TSOs using real network models reveal that there are different implementations by the Suppliers. The present document defines required implementation for some of the CGMES classes/attributes in order or achieve interoperability.

E.2 TapChanger.neutralU vs PowerTransformerEnd.ratedU vs VoltageLevel.BaseVoltage

E.2.1 Issue description

The following questions were addressed to clarify the use of TapChanger.neutralU and PowerTransformerEnd.ratedU:

1) Does RatioTapChanger.stepVoltageIncrement relate to TapChanger.neutralU or PowerTransformerEnd.ratedU?
2) When and why can PowerTransformerEnd.ratedU differ from TapChanger.neutralU?
3) Is TapChanger.neutralU relevant for PhaseTapChanger (PhaseTapChangerAsymmetrical, PhaseTapChangerNonLinear, PhaseTapChangerSymmetrical and PhaseTapChangerLinear)?
4) Is SvTapStep.position mandatory in the SV file when a transformer has TapChanger.controlEnabled = false?

These questions were discussed within ENTSO-E and IEC-TC57-WG13. The discussion on the different items, which are mentioned above, can be summarized as follows:

1)
It should be taken into account that the description is: "Tap step increment, in per cent of nominal voltage, per step position." Please also note that for PhaseTapChanger it is explicitly mentioned that voltageStepIncrement relates to "nominal voltage of the transformer end" = PowerTransformerEnd.ratedU.

Hence, the neutralU discussion is then only relevant for RatioTapChanger.

The advice from WG13 is that on RatioTapChanger.stepVoltageIncrement should be neutral voltage, not nominal. WG13 will write a CIM issue to clarify this in the upcoming standards. On PhaseTapChangerNonLinear.voltageStepIncrement description is currently “nominal voltage of the transformer end”. WG13 proposes to change this to “neutral voltage of the tap changer”.

2)
WG13 advises that TapChanger.neutralU is the voltage that will be at the neutral step. Conventions can differ.

3)
WG13 advises that it is only relevant for PhaseTapChangerNonLinear.

4)

WG13 advises that SvTapStep.position should be mandatory. WG13 will prepare an update of the IEC61970-456.

E.2.2 Required implementation

Considering the outcome of the ENTSO-E discussion, consultations with WG13 and concerned suppliers implementing CGMES, ENTSO-E requires the following implementation:

- TapChanger.neutralU is the voltage at the terminal of the PowerTransformerEnd associated with the tap changer when all tap changers on the transformer are at their neutralStep position. Normally neutralU of the tap changer is the same as ratedU of the PowerTransformerEnd, but it can differ in special cases such as when the tapping mechanism is separate from the winding more common on lower voltage transformers. For CGMES neutralU equals ratedU.
- RatioTapChanger.stepVoltageIncrement shall be in per cent of neutral voltage, per step position, not nominal. The right description of this attribute is: Tap step increment, in per cent of neutral voltage, per step position.
- Nominal quantities are not related to the equipment but to the system nominal voltage in the grid.
- Rated quantities such as ratedU are related to the nameplate data.
- TapChanger.neutralStep is the step position where the voltage is neutralU when the other terminals of the transformer are at the ratedU. If there are other tap changers on the transformer those taps are kept constant at their neutralStep.
- For PhaseTapChangerAsymmetrical, PhaseTapChangerSymmetrical and PhaseTapChangerLinear the neutralU is not relevant.
- PhaseTapChangerNonLinear.voltageStepIncrement relates to the ratedU. The description will be updated by IEC/WG13. The voltageStepIncrement is used for PhaseTapChangerAsymmetrical, PhaseTapChangerSymmetrical.
- SvTapStep.position shall be required even if the TapChanger.controlEnabled is set to false.

As CGMES requires neutralU to be equal to ratedU, the following implementations are valid:

- ratedU + ratedU* stepVoltageIncrement*(SvTapStep.position-neutralStep)
- neutralU + neutralU*stepVoltageIncrement*(SvTapStep.position-neutralStep)
- neutralU + ratedU*stepVoltageIncrement*(SvTapStep.position-neutralStep)
- ratedU + neutralU*stepVoltageIncrement*(SvTapStep.position-neutralStep)

E.3 Angle of PhaseTapChangerTaple Point

PhaseTapChangerTablePoint.angle in CGMES shall be implemented considering that “A positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer).

E.4 Slack generator

The following rules are applied regarding the slack generator:

- SynchronousMachine.referencePriority is used to define the slack generator - reference angle. SynchronousMachine.referencePriority is a required attribute in the SSH.
- In SSH - SynchronousMachine.referencePriority - Priority of unit for use as powerflow voltage phase angle reference bus selection. 0 = don’t care (default) 1 = highest priority. 2 is less than 1 and so on. Please note that GeneratingUnit.normalPF is used for representing distributed slack participation factor.

- In SV - TopologicalIsland. AngleRefTopologicalNode - The angle reference node is the TopologicalNode to which a synchronous machine is connected and referenced from TopologicalIsland.AngleRefTopologicalNode.

Therefore, this is the TopologicalNode to which a SynchronousMachine is connected and has SynchronousMachine.referencePriority equals 1 in SSH.

E.5 qPercent SynchronousMachine

The attribute SynchronousMachine.qPercent shall be used as a participation factor not necessarily summing up to 100% for the participating devices in the control. The reason for this statement is that the attribute is exchanged in the EQ profile. The participation of the devices regulating particular node can be set in other profiles such as SSH. Therefore, the implementations shall consider this in the load flow calculation and normalize the participation based on what devices are participating in the regulation of a particular node.

E.6 TopologicalIsland

The TopologicalIsland-s for a merged model which are defined in the state variables instance file for the merged model are created with the solving on the power flow of the merged model, i.e. there are no TopologicalIsland-s defined per MAS in a merged model. In case a solved model is exchanged for a single MAS the state variables profile must include at least one instance of TopologicalIsland.

E.7 Implementation of SSH and SV profiles

Load sign convention for SSH and SV is used for attributes where the sign convention is not specified. For reactive power (e.g. on RotatingMachine, StaticVarCompensator, etc.) positive means inductive.
For EQ ratings the equipment is used as a reference rather than the node.

E.8 Ground voltage levels

There shall not be any ground voltage level. The ConnectivityNode connected to the Ground instance belongs to the same containment instance as the grounding device. This means that the Ground instances are contained in the same voltage level where other switching devices are contained with a non-zero BaseVoltage.

E.9 LTCflag

E.9.1 Issue description

The problem is related to the attribute TapChanger.ltcFlag (description of the attribute is “Specifies whether or not a TapChanger has load tap changing capabilities.”) and the association TapChanger.TapChangerControl.

Previously agreed rules are the following:

- If TapChanger.ltcFlag is true, and TapChanger.TapChangerControl is not provided, Error: Missing association TapChanger.TapChangerControl, required if attribute TapChanger.ltcFlag is 'true'.
- If TapChanger.ltcFlag is false, and TapChanger.TapChangerControl is provided, Warning: "Association TapChanger.TapChangerControl is ignored, because attribute TapChanger.ltcFlag is 'false'.
- If TapChanger.ltcFlag is false, and TapChanger.TapChangerControl is not provided, Alert: "The TapChanger is fixed."

- If TapChanger.ltcFlag is true, TapChanger.TapChangerControl is provided, and TapChanger.controlEnabled is false, Alert: "The TapChanger control is disabled."
- If TapChanger.ltcFlag is false, and TapChanger.controlEnabled is true, Warning: "The control is enabled for a fixed TapChanger."
- If TapChanger.controlEnabled is true, and SvTapStep is not provided, Error: "SvTapStep is not provided, required if TapChanger.controlEnabled is true."

The above rules were based on the interpretation that:

- TapChanger.ltcFlag is used to indicate whether there is physical tap changer controller installed. For a fixed or a manually controlled transformer, this flag is ‘false’. Then the association to TapChanger.TapChangerControl is not needed.
- TapChanger.controlEnabled is a flag indicating that whether the installed tap changer is actually in-service. You could have an installed tap changer, but not put it into the service. In other words, the tap changer is disabled. In that case, the association to TapChanger.TapChangerControl is still needed. The control scheme is just not enabled.

It was assumed that TapChanger.ltcFlag is not only used for specifying tap change controlling capability. It is not sure whether (voltage/active power regulating) load tap changing (LTC) is different from regular (manual) tap changing. That might be the case as there are no flags similar to TapChanger.ltcFlag in other voltage regulating equipment (ShuntCompensator, etc.). If that is the case, tehre is a need to decouple attribute TapChanger.ltcFlag from association TapChanger.TapChangerControl.

Based on the above rules it is no longer possible to export a transformer with TapChanger.ltcFlag=true and without given TapChanger.TapChangerControl. There are significant number of transformers which are modelled with on-load tap changers which are switched manually. These are currently modelled with on-load tap changing capability, but without controller. Nevertheless, in the past these transformers have been modelled with a minimum voltage and a maximum voltage at the controlled bus section, but with the control enabled flag set to false (TapChanger.controlEnabled=false) in the legacy systems.

The TapChanger.ltcFlag option is also relevant and used for the IEC short-circuit calculation, so vendors cannot reset the option to "false", because this leads to incorrect short-circuit results. Further a creation of a TapChanger.TapChangerControl is also not preferred, because a tap-controller does not exist in reality. It is possible however, but not desirable because the control data would be fake data.

E.9.2 Use cases

The following use cases are defined assuming that different datasets of the instance data that is exchanged with CGMES serve different purpose from operational planning to long term planning cases.

It is also assumed that in case the capabilities of the physical equipment are changing after a point in time the change will be properly reflected in the EQ and SSH profiles or the software enables the user to modify or make different use of this data during specific analyses.

Depending on the settings of the power flow algorithm different software can produce a different power flow resutls, e.g. tap positions may be different, the reactive injections at synchronous machines would be different, the reference generator (slack) active power will differ.

The use cases are defined for the purpose of tap regulation and may not cover all other aspects for operational or long term planning.

E.9.2.1 Power flow calculation relies on SSH

The calculation of a power flow taking EQ and SSH as an input is the most frequent use case for initial calculation. It is mostly used when a party (e.g. a TSO) is requested to provide unsolved model (EQ and SSH, for node- breaker model or EQ, SSH and TP for bus-branch model), or a solved model is to be prepared (EQ, SSH, TP and SV).

In this situation the software makes use of classes and attributes in EQ and SSH to perform the topology process and run power flow calculation. The outcome is TP (from the topology processing function) and SV (from the power flow calculation). Tap regulation could be essential for converging of the power flow algorithm and for achieving good voltage profile. Therefore the power flow algorithm normally provides capabilities to the user to enable or disable tap regulation during power flow. This is normally a global setting which makes use of EQ and SSH information to apply or not apply tap regulation for different transformers.

There is no modification of EQ and SSH data but just use of it. In case when the user needs to perform specific studies (e.g. modification of equipment to enable physical capability to regulate on load or to enable the regulation for a particular tap changer) he/she needs to perform set of actions on the EQ and SSH data so that the software is capable to consider the new situation.

E.9.2.2 Power flow calculation relies on SV

In some cases it is required that the starting point of the power flow algorithm is a previously computed solution. This mainly applies for business processes in which an entity is using the data from multiple parties and providing service to all parties by performing assembling and power flow calculation for the assembled model. The outcome of the service is a single SV instance file representing the solution (e.g. for the ENTSO-E Common Grid Model).

In order to make use of this service individual parties that provided data would rely on the SV to reproduce exactly the same power flow solution. In case the power flow runs from SV the software will make use of tap positions (and other data) that are present in the SV data (as well as the data in EQ and TP).

E.9.2.3 Power flow calculation relies on either SSH or SV

This a combined situation in which the receiving software has access to both SSH and SV data and enables the user to select what kind of data will be the basis for the power flow calculation. In all cases no data is modified but it is just that the software uses one or another set of data.

E.9.3 Required implementation

Control of tap changers in a power flow type of application is made using the TapChangerControl class and the TapChanger.ltcFlag. If a TapChanger has a TapChangerControl (referenced TapChanger.TapChangerControl) means that the power flow application may control the tap changer. The TapChanger.ltcFlag provides information that the TapChanger has physical capability to move the tap under load. Also used in the IEC 60909 calculations to indicate if the tap can move on load.

The meaning of the combinations for TapChanger.TapChangerControl and TapChanger.ltcFlag are described in the following table.

TapChanger.ltcFlag	*TapChanger.TapChangerControl*	*Description*
False	Not present	A real and fixed tap that is not controlled and cannot be moved on load (manual tap change). Power flow cannot be set to change the tap for voltage control during the calculation.
True	Not present	A real and fixed tap that is not controlled but can be moved under load, e.g. manually. Power flow cannot be set to change the tap for voltage control during the calculation. Optimal power flow might have access to these taps and change in order to optimize. Also State Estimator may estimate the tap position to find a better solution for the system state.
True	Present	A real tap with a possibility to change taps automatically for voltage control/active power (load flow) enabled. Depending on the RegulatingControl.enabled and TapChanger.controlEnabled in SSH, the power flow shall or shall not participate in the regulation. In cases where the RegulatingControl is associated with more than one tap changer or other devices the attribute TapChanger.controlEnabled can be set to false in order to set which of the taps are not enabled.

False	Present	An artificial tap changer can be used to simulate control behavior in power flow.

Please note that SvTapStep is required for all TapChangers.

E.10ACLineSegment-s between different terminal voltages

E.10.1Issue description

Different implementations by the Suppliers for lines connecting nodes having different nominal voltage lead to different flows on lines that have same physical characteristics. This happens due to the fact that some implementations are using different base voltage when converting the physical units of the line to per unit system and vice versa. This is especially valid when multiple parties are importing and re-exporting between different tools.

In order to resolve this issue CGMES version 2.4.15 clearly specifies that “Each ACLineSegment is required to have an association to a BaseVoltage. The association to Line is not required. ENTSO-E exchanges allow 10 % difference of the BaseVoltage.nominalVoltage at the two ends of an ACLineSegment representing a complete tie-line or connecting to a boundary node”

OCL constraint on this was also added to CGMES.

E.10.2Required implementation

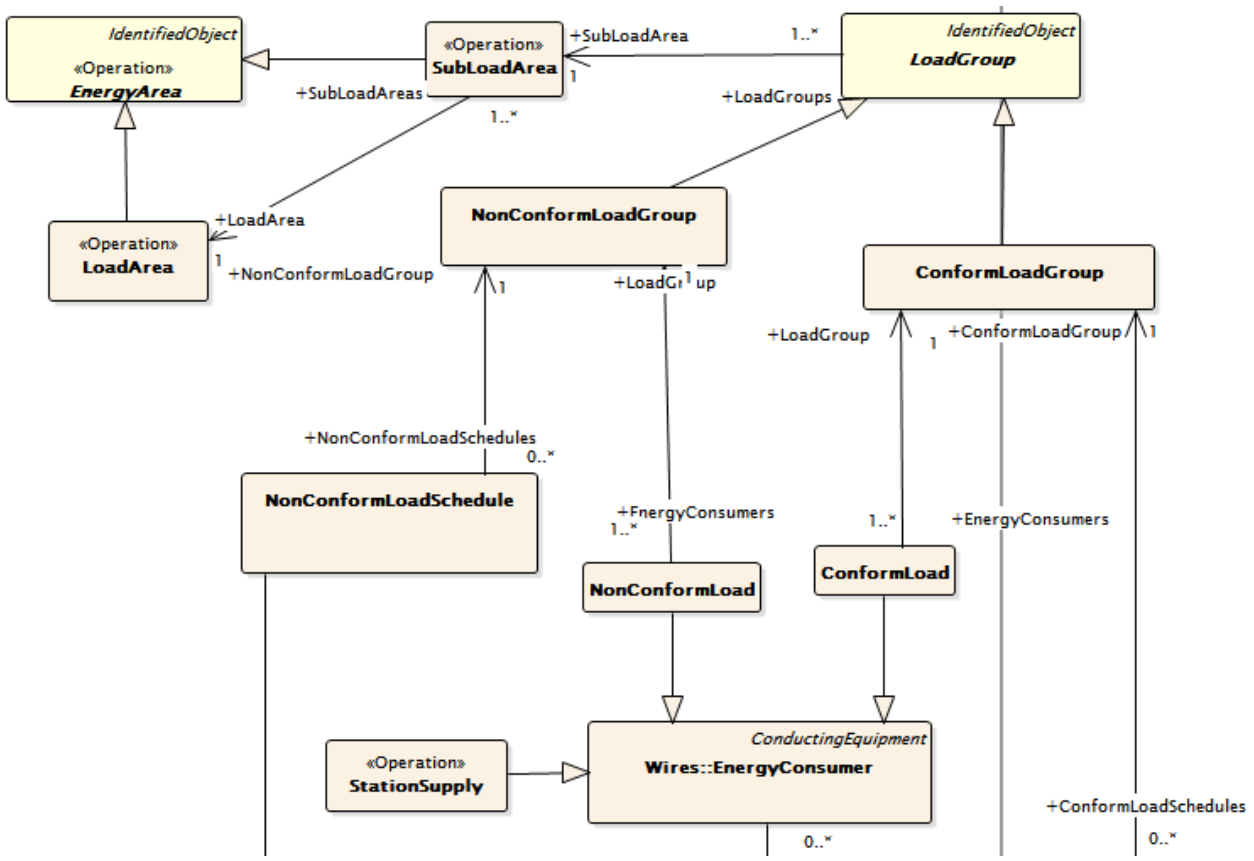
Considering the outcome of the ENTSO-E discussion, consultations with WG13 and concerned suppliers implementing CGMES, ENTSO-E requires the following implementation:

- All implementations shall use association to a BaseVoltage for the purpose of any per unit calculations and shall not rely on the voltages (neither nominal nor actual values obtained by previous or current solution) at the nodes, which the ACLineSegment connects to.
- In case there are interconnected ACLineSegments with different BaseVoltage for different parts of the networks (when assembling different model authority sets) the application needs to handle this to ensure accurate physical units.

E.11Association from ConformLoadGroup/NonConformLoadGroup

E.11.1Issue description

SubLoadArea is marked as “Operation”. Classes ConformLoadGroup and NonConformLoadGroup inherit the association to SubLoadArea from LoadGroup. Therefore, it is logical that the association is also marked with stereotype “Operation”.



E.11.2Required implementation

Considering the outcome of the ENTSO-E discussion, consultations with WG13 and concerned suppliers implementing CGMES, ENTSO-E requires the following implementation:

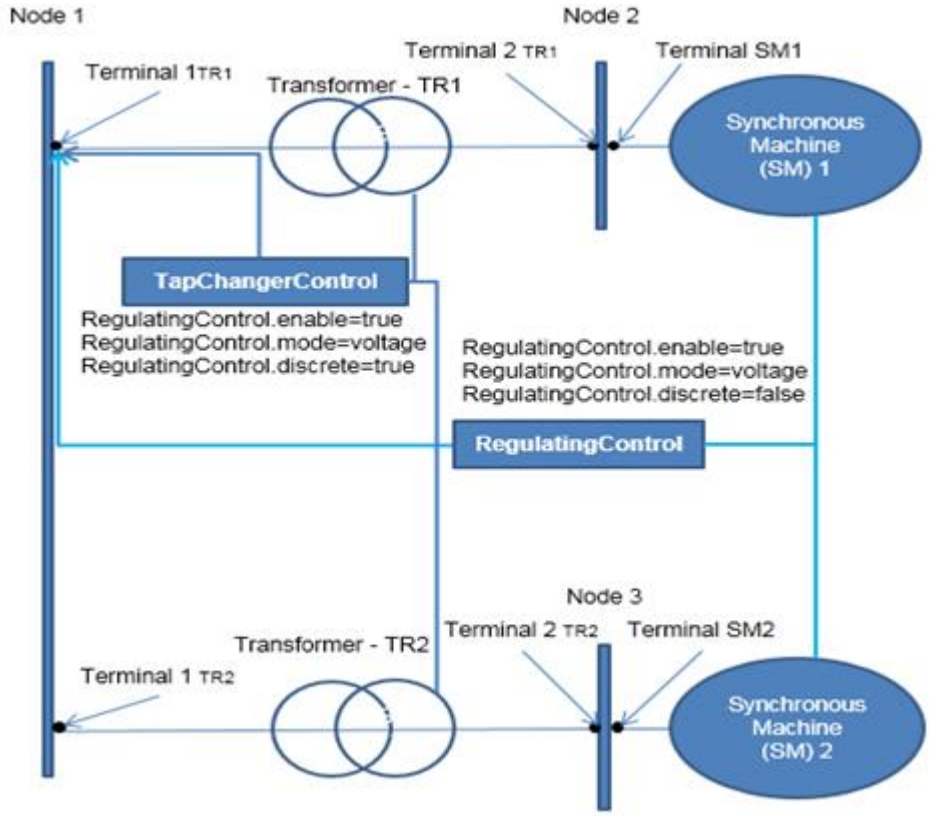
- The association from ConformLoadGroup and NonConformLoadGroup to SubLoadArea is required for “Operation”, but optional for Base.
- The association “SubLoadArea” for ConformLoadGroup and NonConformLoadGroup is considered as “Operation” and will be marked in the next version of the CGMES.

ENTSO-E would like to inform that CIMdesk application have implemented the following validation rules:

- Each ConformLoad and NonConformLoad shall be associated with ConformLoadGroup and NonConformLoadGroup. This is for both “Equipment Core” (Base) and “Operation’ validation.
- Association from ConformLoadGroup and NonConformLoadGroup to SubLoadArea is required for “Operation”, but optional for Base.

E.12 Regulating control

RegulatingControl description specifies that in case multiple equipment, possibly of different types, control same terminal there must be only one RegulatingControl at that terminal. The most specific subtype of RegulatingControl shall be used in case such equipment participate in the control, e.g. TapChangerControl for tap changers. This setup is illustrated in the following figure. The applications shall ensure that the controlled terminal is the one of the controlled node.



E.13 Implementation of the GeographicalRegion ans SubGeographicalRegion

The following best practice implementation is recommended related to GeographicalRegion ans SubGeographicalRegion:

- One GeographicalRegion should be exchanged per MAS. In case TSOs have a need to have the same GeographicalRegion (i.e. multiple TSOs in a country) the class GeographicalRegion shall be present in all TSO models and shall have different rdf:ID, but can have same name/description.
- SubGeographicalRegion is normally a TSO or sub-area of a TSO.
- There is no specific naming convention defined.

E.14 Implementation of GeneratingUnit.normalPF

It is allowed to have values for GeneratingUnit.normalPF which sum is different than 1. The application that imports such model shall have a load flow calculation logic setup (for different MAS that use very different normalPF values, the values must be normalized) to use the information for the purpose of the distributed slack and preserve the original values of GeneratingUnit.normalPF in case of export.

Example of GeneratingUnit.normalPF values (before normalization) for different MAS

	MAS 1	MAS 2	MAS 3
GeneratingUnit.normalPF for Gen 1	0	200	50
GeneratingUnit.normalPF for Gen 2	0.3	500	60

GeneratingUnit.normalPF for Gen 3	0.4	150	45

E.15 Implementation of Power Transformer

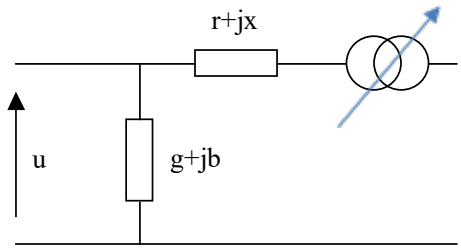
The applications shall follow strictly the description of the PowerTransformerEnd which states “for a two Terminal PowerTransformer the high voltage PowerTransformerEnd has non zero values on r, r0, x, and x0 while the low voltage PowerTransformerEnd has zero values for r, r0, x, and x0.”.

The high voltage side is given by the TransformerEnd.endNumber: “Number for this transformer end, corresponding to the end's order in the power transformer vector group or phase angle clock number. Highest voltage winding should be 1. Each end within a power transformer should have a unique subsequent end number. Note the transformer end number need not match the terminal sequence number.”

In case of a two winding transformer with same rated voltage (PowerTransformerEnd.ratedU) on both sides the application makes sure that only one side has TransformerEnd.endNumber equals to 1.

Therefore the parameters are always provided for PowerTransformerEnd which has TransformerEnd.endNumber equal to 1.

E.16 Interpretation of parameters of PowerTransformerEnd



A two winding PowerTransformer has two PowerTransformerEnds. This gives the option to specify the impedance values for the equivalent pi-model completely at one end or split them between the two ends. The impedances shall be specified at the primary voltage side.

The left side is the “primary” (high voltage) voltage side.

E.17 Add the following to specify the implementation of Switch

There are two different use cases related to the association Terminal.TopologicalNode:

- reduction away of the Switching details to create bus-branch style model intended for traditional planning only.
- keep the detailed model including the Switches and provide topology results also about the Switches. In cases of creation of a bus-branch model from a node-breaker model non-retained Switches are of no interest and their Terminal.TopologicalNode references shall not be included. If instead the model is intended to stay node-breaker it is of interest to know
- if a Switch has been reduced away indicated by the two Switch Terminal.TopologicalNodes referring the same TopologicalNode.
- if an open Switch connect two different TopologicalNodes that is useful when studying the consequences of closing the switch.

In this case Terminal.TopologicalNode shall be included for all Switches.

If the two sides of a Switch are connected to the same ConnectivityNode this should be considered an error.

A Switch with the two sides connected to the same TopologicalNode is valid and normal situation, if not retained.

A Switch connecting BusbarSections at different VoltageLevels is an error.

E.18 UnitMultiplier

UnitMultiplier is not exchanged. It is defined that the multiplier is:

- M for W, VA, VAh, Wh, VArh and Var
- k for volt
- 1 for all the rest of UnitSymbol values.

E.19 EnergySource: “voltageMagnitude” and “voltageAngle”

impedance values for the equivalent

- EnergySource.activePower and EnergySource.reactivePower in the SSH are defining PQ generation. EnergySource.voltageAngle and EnergySource.voltageMagnitude are not exchanged in the EQ.
- EnergySource.voltageAngle and EnergySource.voltageMagnitude are specified as voltage characteristics imposed on the node (at the Terminal).
- EnergySource.voltageAngle and EnergySource.voltageMagnitude are moved to SSH.
- The following use case needs to be documented.
    - This is understood such that a transmission high voltage solution exists and the consequences of that solution is studied on a connected distribution network. The voltages and angles from the solved transmission level solution is then used as input to the distribution network.
    - The solution from the transmission level may have used and equivalent of the distribution network so it is difficult to tell whether that transmission level solution is based on an open or closed circuit representation of the distribution network.

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Annex F
(normative)

CGMES profiles versions

The CGMES version 2.4.15 uses the following profiles and extensions:

- XMI: ENTSOE_CGMES_v2.4.15_7Aug2014_XMI.zip
- HTML: ENTSOE_CGMES_v2.4.15_7Aug2014_HTML.zip
- RDFS: ENTSOE_CGMES_v2.4.15_7Aug2014_RDFS.zip
- HTML_EA: ENTSOE_CGMES_v2.4.15_7Aug2014_HTML_EA.zip
- PDF: ENTSOE_CGMES_v2.4.15_7Aug2014_PDF.pdf
- OCL: ENTSOE_CGMES_v2.4.15_7Aug2014_OCL.zip
- EQ_BD profile version: 3.1
- TP_BD profile version: 3.1
- EQ profile version: 3.1
- TP profile version: 4.1
- SSH profile version: 1.1
- SV profile version: 4.1
- DL profile version: 3.1
- GL profile version: 2.1

Bibliography

[1] Bibliography

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